Integração e Tratamento de Dados - Exercício

IFSP - PEP

Cláudio M. Zaina

Apresentação de técnicas de extração, transformação e carga (*extract*, *transform*, *load* - ETL) de dados utilizando R.

Índice

1 Integração e Tratamento de Dados **1**
1.1 Preparação do ambiente: . . . 1
1.2 Extração . . . 2
1.3 Transformação . . . 7
Limpeza e preparação das variáveis . . . 7
`tempo` . . . 7
`idade` . . . 8
`industria` . . . 9
`cargo` . . . 16
`cargo_adic` . . . 18
`salario` . . . 20
`moeda` . . . 23
`moeda_outra` . . . 24
`pais` . . . 24
`estado_eua` . . . 32
`cidade` . . . 35
`anos_exp_total` . . . 42
`anos_exp_area` . . . 44
`educacao` . . . 46
`sexo` . . . 48
`etnia` . . . 50
1.4 Conformação do salário em USD . . . 55
1.5 Separando observações próprias para análise . . . 59

1 Integração e Tratamento de Dados

1.1 Preparação do ambiente:

Ao iniciar um processo de obtenção e tratamento de dados, é sempre importante manter um padrão de trabalho: quais bibliotecas são carregadas, quais funções são tornadas disponíveis, ajuste dos parâmetros para resultados dentro da norma da empresa etc.

```
rm(list=ls())
```

```r
# biblioteca de definição de representações
library(repr)
# função para definir dimensão da célula de gráfico no Jupyter:
dimgraf <- function (x=11, y=7) {
    options(repr.plot.width=x,repr.plot.height=y)
}
```

1.2 Extração

Obtenção dos dados

```r
sal <- read.csv("SalarySurvey2021.csv", header=TRUE, sep=",")
```

Resumo:

```r
summary(sal)
```

```
 Timestamp         How.old.are.you.   What.industry.do.you.work.in.
Length:28075       Length:28075       Length:28075
Class :character   Class :character   Class :character
Mode  :character   Mode  :character   Mode  :character
```

```
 Job.title
Length:28075
Class :character
Mode  :character
```

```
If.your.job.title.needs.additional.context..please.clarify.here.
Length:28075
Class :character
Mode  :character
```

```
What.is.your.annual.salary...You.ll.indicate.the.currency.in.a.later.question..If.you.are.part.time.or.ho
Length:28075
Class :character
Mode  :character
```

```
How.much.additional.monetary.compensation.do.you.get..if.any..for.example..bonuses.or.overtime.in.an.aver
Min.   :        0
1st Qu.:        0
Median :     2000
Mean   :    18159
3rd Qu.:    10000
```

```
Max.   :120000000
NA's   :7302
Please.indicate.the.currency If..Other...please.indicate.the.currency.here..
Length:28075                 Length:28075
Class :character             Class :character
Mode  :character             Mode  :character
```

```
If.your.income.needs.additional.context..please.provide.it.here.
Length:28075
Class :character
Mode  :character
```

```
What.country.do.you.work.in. If.you.re.in.the.U.S...what.state.do.you.work.in.
Length:28075                 Length:28075
Class :character             Class :character
Mode  :character             Mode  :character
```

```
What.city.do.you.work.in.
Length:28075
Class :character
Mode  :character
```

```
How.many.years.of.professional.work.experience.do.you.have.overall.
Length:28075
Class :character
Mode  :character
```

```
How.many.years.of.professional.work.experience.do.you.have.in.your.field.
Length:28075
Class :character
Mode  :character
```

```
What.is.your.highest.level.of.education.completed. What.is.your.gender.
Length:28075                                       Length:28075
Class :character                                   Class :character
Mode  :character                                   Mode  :character
```

```
 What.is.your.race...Choose.all.that.apply..
 Length:28075
 Class :character
 Mode  :character
```

O arquivo de dados `CSV` contém, como nome das variáveis, o texto da questão. É necessário então nomear as variáveis segundo seu significado.

```
str(sal)
```

```
'data.frame':   28075 obs. of  18 variables:
 $ Timestamp
 $ How.old.are.you.
34" "25-34" "25-34" "25-34" ...
 $ What.industry.do.you.work.in.
 $ Job.title
 $ If.your.job.title.needs.additional.context..please.clarify.here.
 $ What.is.your.annual.salary...You.ll.indicate.the.currency.in.a.later.question..If.you.are.part.time.or.
 $ How.much.additional.monetary.compensation.do.you.get..if.any..for.example..bonuses.or.overtime.in.an.av
 $ Please.indicate.the.currency
 $ If..Other...please.indicate.the.currency.here..
 $ If.your.income.needs.additional.context..please.provide.it.here.
 $ What.country.do.you.work.in.
 $ If.you.re.in.the.U.S...what.state.do.you.work.in.
 $ What.city.do.you.work.in.
 $ How.many.years.of.professional.work.experience.do.you.have.overall.
7 years" "8 - 10 years" "2 - 4 years" "8 - 10 years" ...
 $ How.many.years.of.professional.work.experience.do.you.have.in.your.field.
7 years" "5-7 years" "2 - 4 years" "5-7 years" ...
 $ What.is.your.highest.level.of.education.completed.
 $ What.is.your.gender.
binary" "Woman" "Woman" ...
 $ What.is.your.race...Choose.all.that.apply..
```

- "timestamp" → `tempo`;
- "How.old.are.you" → `idade`
- "What.industry.do.you.work.in" → ramo de trabalho, `industria`;
- "Job.title" → nome do `cargo`;
- "If.your.job.title.needs.additional.context..please.clarify.here" → explicações adicionais do cargo (`cargo_adic`);
- "What is your annual salary. You'll indicate the currency in a later question. If you are part time or hourly please enter an annualized equivalent. what you would earn if you worked the job 40 hours a week 52 weeks a year → `salario`;
- "How much additional monetary compensation do you get if any for example bonuses or overtime in an average year Please only include monetary compensation here not the value of benefits" → bônus ou recebimento adicional (`bonus`);
- "Please.indicate.the.currency" → `moeda`;
- "If..Other…please.indicate.the.currency.here" → se outra moeda (`moeda_outra`);
- "If.your.income.needs.additional.context..please.provide.it.here." → se salário possui contexto adicional (`salario_contexto`);
- "What.country.do.you.work.in" → `pais`;

- "If.you.re.in.the.U.S...what.state.do.you.work.in" → se nos EUA, qual estado (`estado_eua`);
- "What.city.do.you.work.in" → `cidade`;
- "How.many.years.of.professional.work.experience.do.you.have.overall" → `anos_exp_total`;
- "How.many.years.of.professional.work.experience.do.you.have.in.your.field" → `anos_exp_area`;
- "What.is.your.highest.level.of.education.completed" → nível de educação (`educacao`);
- "What.is.your.gender" → `sexo`;
- "What.is.your.race...Choose.all.that.apply" → "raça" (`etnia`).

```
names(sal) <- c("tempo", "idade", "industria", "cargo", "cargo_adic", "salario", "bonus",
                "moeda", "moeda_outra","salario_contexto", "pais", "estado_eua", "cidade",
                "anos_exp_total", "anos_exp_area", "educacao", "sexo", "etnia")
```

```
summary(sal)
```

```
   tempo              idade            industria            cargo          
Length:28075       Length:28075       Length:28075       Length:28075      
Class :character   Class :character   Class :character   Class :character  
Mode  :character   Mode  :character   Mode  :character   Mode  :character  
 cargo_adic          salario              bonus              moeda          
Length:28075       Length:28075       Min.   :        0   Length:28075      
Class :character   Class :character   1st Qu.:        0   Class :character  
Mode  :character   Mode  :character   Median :     2000   Mode  :character  
                                      Mean   :    18159                     
                                      3rd Qu.:    10000                     
                                      Max.   :120000000                     
                                      NA's   :7302                          
moeda_outra        salario_contexto       pais            estado_eua        
Length:28075       Length:28075       Length:28075       Length:28075      
Class :character   Class :character   Class :character   Class :character  
Mode  :character   Mode  :character   Mode  :character   Mode  :character  
   cidade          anos_exp_total     anos_exp_area        educacao         
Length:28075       Length:28075       Length:28075       Length:28075      
Class :character   Class :character   Class :character   Class :character  
Mode  :character   Mode  :character   Mode  :character   Mode  :character  
    sexo               etnia          
Length:28075       Length:28075       
Class :character   Class :character   
Mode  :character   Mode  :character   
```

Qual o tipo do conteúdo das variáveis?

```
print(head(sal))
```

```
               tempo idade                     industria
1 4/27/2021 11:02:10 25-34  Education (Higher Education)
2 4/27/2021 11:02:22 25-34              Computing or Tech
3 4/27/2021 11:02:38 25-34 Accounting, Banking & Finance
4 4/27/2021 11:02:41 25-34                    Nonprofits
5 4/27/2021 11:02:42 25-34 Accounting, Banking & Finance
6 4/27/2021 11:02:46 25-34  Education (Higher Education)
                                   cargo cargo_adic salario bonus moeda
1       Research and Instruction Librarian             55,000     0   USD
2 Change & Internal Communications Manager             54,600  4000   GBP
3                     Marketing Specialist             34,000    NA   USD
4                          Program Manager             62,000  3000   USD
5                        Accounting Manager             60,000  7000   USD
6           Scholarly Publishing Librarian             62,000    NA   USD
  moeda_outra salario_contexto           pais     estado_eua      cidade
1                              United States  Massachusetts      Boston
2                             United Kingdom                  Cambridge
3                                         US      Tennessee Chattanooga
4                                        USA      Wisconsin   Milwaukee
5                                         US South Carolina  Greenville
6                                        USA  New Hampshire     Hanover
  anos_exp_total anos_exp_area        educacao       sexo etnia
1      5-7 years     5-7 years Master's degree      Woman White
2   8 - 10 years     5-7 years  College degree Non-binary White
3    2 - 4 years   2 - 4 years  College degree      Woman White
4   8 - 10 years     5-7 years  College degree      Woman White
5   8 - 10 years     5-7 years  College degree      Woman White
6   8 - 10 years   2 - 4 years Master's degree        Man White
```

```
print(tail(sal))
```

```
                   tempo idade                     industria
28070 9/10/2024 23:58:45 35-44              Computing or Tech
28071 9/12/2024 20:59:43 25-34  Education (Higher Education)
28072 9/13/2024 16:59:35 25-34 Accounting, Banking & Finance
28073 9/17/2024 11:10:17 35-44 Education (Primary/Secondary)
28074 9/17/2024 11:43:33 18-24              Computing or Tech
28075 9/17/2024 12:03:31 18-24              Computing or Tech
                         cargo cargo_adic salario bonus moeda moeda_outra
28070 Business Systems Analyst               90000  9000   USD
28071       Community Director               37741    NA   USD
28072         Staff Accountant               62500    NA   USD
28073         HOMEROOM TEACHER               70000 15000   USD
28074                  Student                 500    NA   USD
28075                  Student                1000   200   USD
                           salario_contexto          pais estado_eua
28070 10% of the salary if te goals are met           USA     Nevada
28071                                       United States    Georgia
28072                                                USA    Georgia
28073                                              China
28074                                       United States    Georgia
28075                                       United States    Georgia
             cidade anos_exp_total  anos_exp_area        educacao  sexo
```

```
28070     Las Vegas   8 - 10 years    2 - 4 years Master's degree Woman
28071 Milledgeville    2 - 4 years 1 year or less Master's degree   Man
28072   Johns Creek 1 year or less 1 year or less Master's degree   Man
28073      Shenzhen   8 - 10 years      5-7 years Master's degree Woman
28074        Tucker 1 year or less 1 year or less  College degree   Man
28075       Roswell 1 year or less 1 year or less    Some college   Man
                          etnia
28070   Asian or Asian American
28071 Black or African American
28072                     White
28073 Black or African American
28074                     White
28075                     White
```

```
save(sal, file="SalarySurvey2021_01_leitura.RData")
```

1.3 Transformação

O próximo passo é a avaliação da estrutura das variáveis e sua adequação para os procedimentos de análise.

```
rm(list=ls()) # Limpando variáveis intermediárias da memória
```

```
load("SalarySurvey2021_01_leitura.RData")
dimgraf <- function (x=11, y=7) { options(repr.plot.width=x,repr.plot.height=y) }
ls()
```

1. 'dimgraf'
2. 'sal'

Limpeza e preparação das variáveis

A maior parte das variáveis é categórica (*fatores*). O marco temporal pode ser convertido via POSIX.

tempo

```
head(sal$tempo)
```

1. '4/27/2021 11:02:10'
2. '4/27/2021 11:02:22'
3. '4/27/2021 11:02:38'
4. '4/27/2021 11:02:41'
5. '4/27/2021 11:02:42'
6. '4/27/2021 11:02:46'

É um registro tipo POSIX do tempo. Pode ser convertido.

```
sal$tempo <- as.POSIXct(sal$tempo, format='%m/%d/%Y %H:%M:%S')
```

```
str(sal$tempo)
```

```
 POSIXct[1:28075], format: "2021-04-27 11:02:10" "2021-04-27 11:02:22" "2021-04-27 11:02:38" ...
```

```
summary(sal$tempo)
```

```
                     Min.                   1st Qu.
"2021-04-27 11:02:10.0000" "2021-04-27 14:21:35.5000"
                   Median                      Mean
"2021-04-28 13:57:11.0000" "2021-05-15 10:18:53.2120"
                  3rd Qu.                      Max.
"2021-04-29 20:31:53.5000" "2024-09-17 12:03:31.0000"
```

idade

```
head(sal$idade)
tail(sal$idade)
```

1. '25-34'
2. '25-34'
3. '25-34'
4. '25-34'
5. '25-34'
6. '25-34'

1. '35-44'
2. '25-34'
3. '25-34'
4. '35-44'
5. '18-24'
6. '18-24'

São faixas etárias. Quais são?

```
unique(sal$idade)
```

1. '25-34'
2. '45-54'
3. '35-44'
4. '18-24'
5. '65 or over'
6. '55-64'
7. 'under 18'

Fator em R:

Seria interessante traduzir as categorias e colocá-las em ordem.

```
sal$idade[sal$idade == 'under 18'] <- '-18'
```

```
sal$idade[sal$idade == '65 or over'] <- '65+'
```

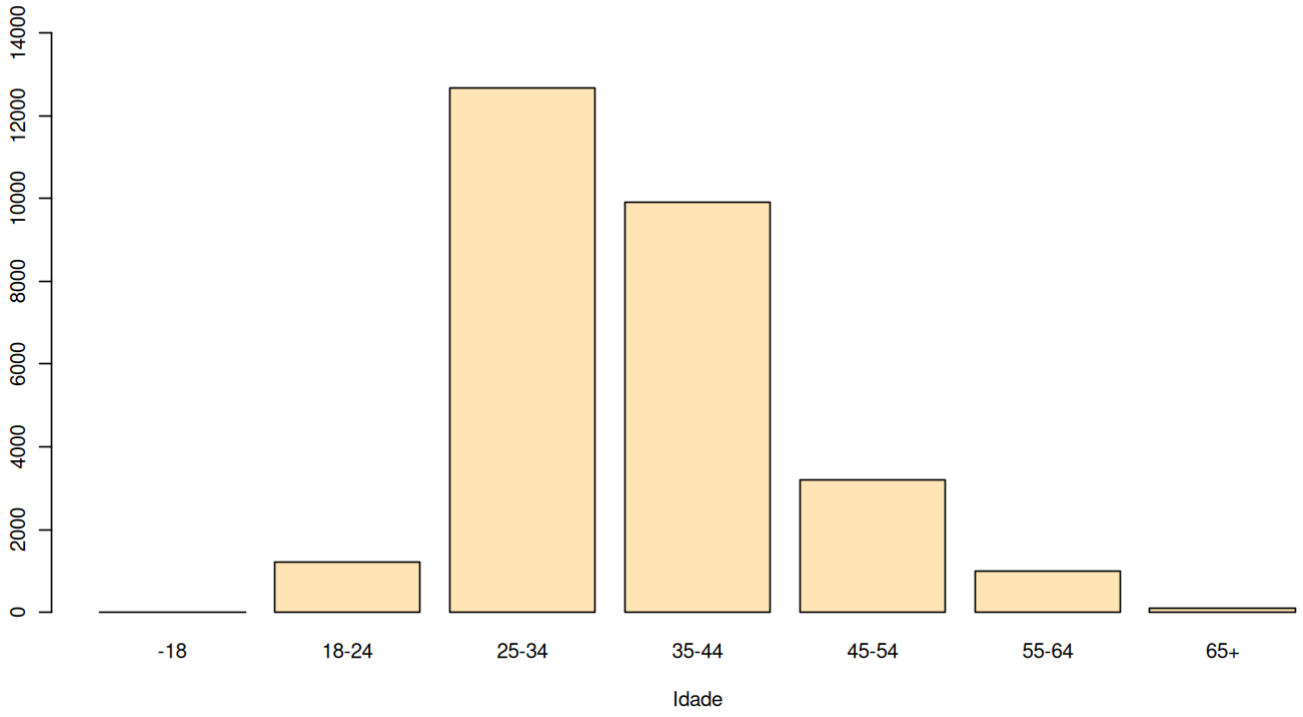
```
idade_ordem <- c("-18", "18-24", "25-34", "35-44", "45-54", "55-64", "65+")
```

```
sal$idade <- factor(sal$idade, ordered=TRUE, levels=idade_ordem, labels=idade_ordem)
```

```
print(summary(sal$idade))
```

```
  -18 18-24 25-34 35-44 45-54 55-64   65+
   14  1224 12660  9902  3189   992    94
```

```
dimgraf(x=12,y=7)
barplot(table(sal$idade),ylim=c(0,15000), col="moccasin", xlab="Idade")
```



industria

```
ind <- tolower(sal$industria)
```

```
head(ind)
tail(ind)
```

1. 'education (higher education)'
2. 'computing or tech'
3. 'accounting, banking & finance'
4. 'nonprofits'
5. 'accounting, banking & finance'
6. 'education (higher education)'

1. 'computing or tech'
2. 'education (higher education)'
3. 'accounting, banking & finance'
4. 'education (primary/secondary)'
5. 'computing or tech'
6. 'computing or tech'

```
print(unique(ind)[1:100])
```

```
  [1] "education (higher education)"
  [2] "computing or tech"
  [3] "accounting, banking & finance"
  [4] "nonprofits"
  [5] "publishing"
```

```
 [6] "education (primary/secondary)"
 [7] "law"
 [8] "health care"
 [9] "utilities & telecommunications"
[10] "business or consulting"
[11] "art & design"
[12] "government and public administration"
[13] "public library"
[14] "engineering or manufacturing"
[15] "media & digital"
[16] "marketing, advertising & pr"
[17] "retail"
[18] "property or construction"
[19] "biotechnology"
[20] "aerospace contracting"
[21] "insurance"
[22] "sales"
[23] "energy"
[24] "environmental regulation"
[25] "hospitality & events"
[26] "transport or logistics"
[27] "medical devices"
[28] "academic research (psychology)"
[29] "social work"
[30] "surveying "
[31] "recruitment or hr"
[32] "phd"
[33] "biopharma"
[34] "stem research"
[35] "libraries"
[36] "architecture"
[37] "academic medicine "
[38] "commercial real estate "
[39] "pet care industry (dog training/walking)"
[40] "politics"
[41] "university administration"
[42] "animal health product manufacturing"
[43] "educational technology - hybrid between book publishing and technology really"
[44] "pharmaceuticals"
[45] "agriculture or forestry"
[46] "philanthropy"
[47] "real estate"
[48] "consulting"
[49] "environmental consulting"
[50] "archaeology / cultural resource management"
[51] "museum"
[52] "biotech"
[53] ""
[54] "\"government relations\" (lobbying) "
[55] "software as a service saas"
[56] "public health in higher education"
[57] "business services"
[58] "law enforcement & security"
[59] "manufacturing and distributing"
[60] "leisure, sport & tourism"
[61] "food manufacturing"
[62] "research and development"
```

```
 [63] "consumer packaged goods"
 [64] "entertainment"
 [65] "b2b services "
 [66] "oil and gas"
 [67] "heritage/public history"
 [68] "manufacturing"
 [69] "veterinary medicine"
 [70] "food production/processing"
 [71] "health insurance "
 [72] "government contractor"
 [73] "public health "
 [74] "renewable energy"
 [75] "aerospace"
 [76] "manufacturing "
 [77] "public library (technically city govt.?) "
 [78] "space"
 [79] "librarian "
 [80] "food service"
 [81] "pharma"
 [82] "aerospace/defense"
 [83] "management consulting"
 [84] "pharmaceutical"
 [85] "pharmaceutical "
 [86] "academic publishing"
 [87] "gas & oil"
 [88] "defense contracting"
 [89] "international development (multilateral donor)"
 [90] "project management "
 [91] "apparel manufacture"
 [92] "auto repair"
 [93] "life sciences"
 [94] "pharma / medical device design and manufacturing"
 [95] "chemical manufacturing"
 [96] "editor in educational publishing"
 [97] "trade association/membership"
 [98] "work-study"
 [99] "science"
[100] "learning & development"
```

```
ind <- factor(ind)
```

```
print(summary(ind))
```

```
                  computing or tech         education (higher education)
                               4704                                 2466
                         nonprofits                          health care
                               2419                                 1896
government and public administration        accounting, banking & finance
                               1890                                 1811
        engineering or manufacturing          marketing, advertising & pr
                               1695                                 1134
                                law               business or consulting
                               1097                                  852
       education (primary/secondary)                       media & digital
                                836                                  774
                          insurance                               retail
```

531	505
recruitment or hr	property or construction
460	387
art & design	utilities & telecommunications
359	357
transport or logistics	sales
304	286
social work	hospitality & events
273	261
entertainment	agriculture or forestry
252	138
leisure, sport & tourism	
98	74
publishing	library
59	56
biotech	libraries
51	51
public library	law enforcement & security
51	43
manufacturing	research
34	34
pharmaceuticals	real estate
26	25
public libraries	architecture
23	22
pharmaceutical	scientific research
21	20
pharma	biotechnology
18	16
energy	environmental consulting
16	16
manufacturing	translation
14	14
market research	oil & gas
13	13
oil and gas	science
13	13
biomedical research	biotech/pharma
12	11
environmental	pharmaceutical
10	10
politics	veterinary
10	10
automotive	museum
9	9
philanthropy	public health
9	9
construction	mining
8	8
publishing	video games
8	8
biotechnology	consulting
7	7
consumer goods	environmental
7	7
museums	software
7	7

```
                academic research                        architecture
                                6                                   6
                  book publishing                      communications
                                6                                   6
                          defense                             fitness
                                6                                   6
                     food service               international development
                                6                                   6
                   medical research                      renewable energy
                                6                                   6
                trade association                    academic publishing
                                6                                   5
                         cannabis                      clinical research
                                5                                   5
         commercial real estate                          construction
                                5                                   5
        consumer packaged goods                      customer service
                                5                                   5
                     e-commerce                             finance
                                5                                   5
         government contracting                   government contractor
                                5                                   5
                             it                           labor union
                                5                                   5
                       library                          life sciences
                                5                                   5
               pharmaceuticals                           public library
                                5                                   5
                       research                             (Other)
                                5                                1225
```

Há muitas categorias. Talvez seja interessante avaliar apenas as mais frequentes. Vamos considerar as categorias que commpreendam 80% das observações:

```
nrow(sal) * 0.8
```

22460

Vamos manter as categorias até 22460 e unificar em `outras` as demais, mesmo sob o custo de alguma categoria estar duplicada com nome diferente.

```
4704 + 2466 + 2419 + 1896 + 1890 + 1811 + 1695 + 1134 + 1097 + 852 +
836 + 774 + 531 + 505 + 460
# 460 foi mantido por estar perto de 505 e longe da próxima 387
```

23070

As categorias que ficaram então são:

```
computing or tech     4704
education (higher education)     2466
nonprofits     2419
health care    1896
government and public administration    1890
accounting, banking & finance    1811
engineering or manufacturing    1695
```

```
marketing, advertising & pr    1134
law    1097
business or consulting    852
education (primary/secondary)    836
media & digital    774
insurance    531
retail    505
```

TODO [fazer análise fina das categorias, unificando nomes similares]
Exemplo:

ind[substr(ind,1,7)==“academi”] <- “academia” ind <- factor(ind) print(sort(levels(ind)))

```
ind <- tolower(sal$industria)
ind[ind != 'computing or tech' &
    ind != 'education (higher education)' &
    ind != 'nonprofits' &
    ind != 'health care' &
    ind != 'government and public administration' &
    ind != 'accounting, banking & finance' &
    ind != 'engineering or manufacturing' &
    ind != 'marketing, advertising & pr' &
    ind != 'law' &
    ind != 'business or consulting' &
    ind != 'education (primary/secondary)' &
    ind != 'media & digital' &
    ind != 'insurance' &
    ind != 'retail'] <- 'other'
ind <- factor(ind)
```

```
print(summary(ind))
```

```
       accounting, banking & finance               business or consulting
                                1811                                  852
                    computing or tech         education (higher education)
                                4704                                 2466
       education (primary/secondary)         engineering or manufacturing
                                 836                                 1695
government and public administration                          health care
                                1890                                 1896
                           insurance                                  law
                                 531                                 1097
         marketing, advertising & pr                        media & digital
                                1134                                  774
                           nonprofits                                other
                                2419                                 5465
                              retail
                                 505
```

```
print(head(sal$industria))
print(head(ind))
```

```
[1] "Education (Higher Education)"  "Computing or Tech"
[3] "Accounting, Banking & Finance" "Nonprofits"
[5] "Accounting, Banking & Finance" "Education (Higher Education)"
```

```
[1] education (higher education)  computing or tech
[3] accounting, banking & finance nonprofits
[5] accounting, banking & finance education (higher education)
15 Levels: accounting, banking & finance ... retail
```

```r
sal$industria <- ind
summary(sal)
```

```
     tempo                          idade
 Min.   :2021-04-27 11:02:10.00   -18  :   14
 1st Qu.:2021-04-27 14:21:35.50   18-24: 1224
 Median :2021-04-28 13:57:11.00   25-34:12660
 Mean   :2021-05-15 10:18:53.21   35-44: 9902
 3rd Qu.:2021-04-29 20:31:53.50   45-54: 3189
 Max.   :2024-09-17 12:03:31.00   55-64:  992
                                  65+  :   94
                                industria       cargo
 other                               :5465   Length:28075
 computing or tech                   :4704   Class :character
 education (higher education)        :2466   Mode  :character
 nonprofits                          :2419
 health care                         :1896
 government and public administration:1890
 (Other)                             :9235
  cargo_adic          salario              bonus              moeda
 Length:28075       Length:28075       Min.   :        0   Length:28075
 Class :character   Class :character   1st Qu.:        0   Class :character
 Mode  :character   Mode  :character   Median :     2000   Mode  :character
                                       Mean   :    18159
                                       3rd Qu.:    10000
                                       Max.   :120000000
                                       NA's   :7302
 moeda_outra        salario_contexto       pais            estado_eua
 Length:28075       Length:28075       Length:28075       Length:28075
 Class :character   Class :character   Class :character   Class :character
 Mode  :character   Mode  :character   Mode  :character   Mode  :character



    cidade          anos_exp_total     anos_exp_area        educacao
 Length:28075       Length:28075       Length:28075       Length:28075
 Class :character   Class :character   Class :character   Class :character
 Mode  :character   Mode  :character   Mode  :character   Mode  :character



     sexo              etnia
 Length:28075       Length:28075
 Class :character   Class :character
 Mode  :character   Mode  :character
```

cargo

```
cargo <- sal$cargo
```

```
head(cargo)
```

1. 'Research and Instruction Librarian'
2. 'Change & Internal Communications Manager'
3. 'Marketing Specialist'
4. 'Program Manager'
5. 'Accounting Manager'
6. 'Scholarly Publishing Librarian'

```
print(summary(factor(cargo)))
```

```
          Software Engineer                Project Manager
                        286                            230
                   Director      Senior Software Engineer
                        198                            196
            Program Manager                        Teacher
                        152                            151
                  Librarian                        Manager
                        150                            145
        Executive Assistant                Product Manager
                        136                            125
  Administrative Assistant                   Data Analyst
                        103                             99
                   Attorney                     Associate
                         97                             94
          Marketing Manager                       Analyst
                         91                             89
             Office Manager             Software Developer
                         89                             89
        Engineering Manager                      Paralegal
                         88                             81
           Business Analyst                     Consultant
                         77                             76
         Operations Manager                     HR Manager
                         76                             75
        Assistant Professor                         Editor
                         73                             71
         Executive Director                 Senior Manager
                         68                             62
         Associate Attorney                 Data Scientist
                         60                             59
            Account Manager             Associate Director
                         57                             57
         Assistant Director           Executive Assistant
                         52                             52
          Software engineer                 Vice President
                         52                             52
        Program Coordinator            Project Coordinator
                         51                             48
              HR Generalist                 Senior Analyst
                         46                             46
                 Controller              Senior Consultant
```

45	45
Associate Professor	Graphic Designer
43	43
Partner	Senior Project Manager
43	42
Accounting Manager	Director of Operations
40	40
Registered Nurse	RN
40	40
Technical Writer	Senior Associate
40	39
Communications Manager	Senior Engineer
38	38
Director of Engineering	Senior Director
37	37
Staff Accountant	Customer Success Manager
37	36
Legal Assistant	Marketing Director
36	36
Research Associate	Staff Software Engineer
36	36
Accountant	Financial Analyst
34	34
Human Resources Manager	Project Engineer
33	33
IT Manager	Program Director
32	32
Senior Editor	Archivist
32	31
Communications Specialist	Scientist
31	31
Senior Product Manager	Software Engineer II
31	31
Chief of Staff	Principal Software Engineer
30	30
Project manager	Senior Software Developer
30	30
Coordinator	Marketing Coordinator
29	29
Professor	Senior Accountant
29	29
Academic Advisor	General Manager
28	28
Instructional Designer	Research Analyst
28	27
Senior Program Manager	Copywriter
27	26
Library Assistant	Mechanical Engineer
26	26
Research Scientist	Supervisor
26	26
Systems Engineer	Administrator
26	25
Engineer	Policy Analyst
25	25
Program manager	Senior Financial Analyst
25	25

```
Senior software engineer                    (Other)
                      25                      22244
```

Os cargos de uma indústria são comuns?

```
head(sal$cargo[sal$industria=='computing or tech'],20)
```

1. 'Change & Internal Communications Manager'
2. 'Systems Analyst'
3. 'Principal Software Engineer'
4. 'Intelligence Analyst'
5. 'Mobile developer'
6. 'Product Design Director'
7. 'Managed Services Sales Coordinator'
8. 'Technical specialist'
9. 'Tech Writer'
10. 'Senior Business Analyst'
11. 'Director of Engineering'
12. 'senior web application developer'
13. 'Chief Engineer'
14. 'Senior Financial Analyst'
15. 'Software engineer'
16. 'Product Manager'
17. 'Service Designer'
18. 'Software Development Manager'
19. 'technical support specialist'
20. 'Systems Analyst'

Não. Aparentemente a indústria tem respondentes de todo tipo de cargo.

Seria interessante manter a variável, mas dificilmente será útil na análise.

cargo_adic

```
head(sal$cargo_adic,20)
tail(sal$cargo_adic,20)
```

1. ''
2. ''
3. ''
4. ''
5. ''
6. ''
7. ''
8. 'High school, FT'
9. 'Data developer/ETL Developer'
10. ''
11. ''
12. ''
13. ''
14. 'non-profit law firm'
15. ''
16. ''
17. ''
18. ''
19. ''
20. ''

1. ''
2. ''
3. ''
4. ''
5. ''
6. ''
7. ''
8. ''
9. ''
10. ''
11. 'Claims'
12. ''
13. ''
14. ''
15. ''
16. ''
17. ''
18. ''
19. ''
20. ''

```
sal$cargo_adic <- NULL
```

```
print(summary(sal))
```

```
    tempo                         idade
 Min.   :2021-04-27 11:02:10.00   -18  :   14
 1st Qu.:2021-04-27 14:21:35.50   18-24: 1224
 Median :2021-04-28 13:57:11.00   25-34:12660
 Mean   :2021-05-15 10:18:53.21   35-44: 9902
 3rd Qu.:2021-04-29 20:31:53.50   45-54: 3189
 Max.   :2024-09-17 12:03:31.00   55-64:  992
                                  65+  :   94
                               industria       cargo
 other                              :5465   Length:28075
 computing or tech                  :4704   Class :character
 education (higher education)       :2466   Mode  :character
 nonprofits                         :2419
 health care                        :1896
 government and public administration:1890
 (Other)                            :9235
   salario              bonus              moeda           moeda_outra
 Length:28075       Min.   :        0   Length:28075       Length:28075
 Class :character   1st Qu.:        0   Class :character   Class :character
 Mode  :character   Median :     2000   Mode  :character   Mode  :character
                    Mean   :    18159
                    3rd Qu.:    10000
                    Max.   :120000000
                    NA's   :7302
 salario_contexto       pais            estado_eua           cidade
 Length:28075       Length:28075       Length:28075       Length:28075
 Class :character   Class :character   Class :character   Class :character
 Mode  :character   Mode  :character   Mode  :character   Mode  :character
```

```
 anos_exp_total     anos_exp_area        educacao             sexo
 Length:28075       Length:28075       Length:28075       Length:28075
 Class :character   Class :character   Class :character   Class :character
 Mode  :character   Mode  :character   Mode  :character   Mode  :character
```

```
   etnia
 Length:28075
 Class :character
 Mode  :character
```

salario

```
head(sal$salario)
tail(sal$salario)
max(nchar(sal$salario))
```

1. '55,000'
2. '54,600'
3. '34,000'
4. '62,000'
5. '60,000'
6. '62,000'

1. '90000'
2. '37741'
3. '62500'
4. '70000'
5. '500'
6. '1000'

11

```
sal$salario[nchar(sal$salario) > 7]
```

1. '1,080,000'
2. '1,150,000'
3. '3,000,000'
4. '1,400,000'
5. '102,000,000'
6. '1,150,000'
7. '1,100,000'
8. '2,800,000'
9. '1,100,000'
10. '1,200,000'
11. '1,900,000'
12. '1,650,000'
13. '1,334,782'
14. '4,600,000'
15. '8,100,000'

16. ‘6,000,000’
17. ‘4,800,000’
18. ‘6,000,000’
19. ‘1,150,000’
20. ‘11,500,000’
21. ‘1,100,000’
22. ‘2,100,000’
23. ‘1,200,000’
24. ‘27,000,000’
25. ‘5,600,000’
26. ‘3,225,000’
27. ‘870,000,000’
28. ‘1,324,000’
29. ‘2,500,000’
30. ‘6,000,000’
31. ‘2,300,000’
32. ‘2,111,538’
33. ‘3,600,000’
34. ‘1,100,000’
35. ‘3,600,000’
36. ‘1,300,000’
37. ‘1,260,000’
38. ‘8,800,000’
39. ‘38,000,000’
40. ‘4,000,000’
41. ‘1,360,000’
42. ‘1,300,000’
43. ‘180,000,000’
44. ‘4,600,000’
45. ‘4,160,000’
46. ‘1,050,000’
47. ‘36000000’
48. ‘43800000’
49. ‘20800000’
50. ‘120000000’
51. ‘10000000’
52. ‘6000070000’

Inicialmente, seria necessário retirar as `,`:

```
sal$salario <- gsub(',', '', sal$salario)
sal$salario[nchar(sal$salario) > 7]
```

1. ‘102000000’
2. ‘11500000’
3. ‘27000000’
4. ‘870000000’
5. ‘38000000’
6. ‘180000000’
7. ‘36000000’
8. ‘43800000’
9. ‘20800000’
10. ‘120000000’
11. ‘10000000’
12. ‘6000070000’

Aparentemente `6000070000` deve ter sido entrado 60000-70000. Substituindo por 65000.

```
sal$salario[sal$salario=='6000070000'] <- '65000'
sal$salario[nchar(sal$salario) > 7]
```

1. '102000000'
2. '11500000'
3. '27000000'
4. '870000000'
5. '38000000'
6. '180000000'
7. '36000000'
8. '43800000'
9. '20800000'
10. '120000000'
11. '10000000'

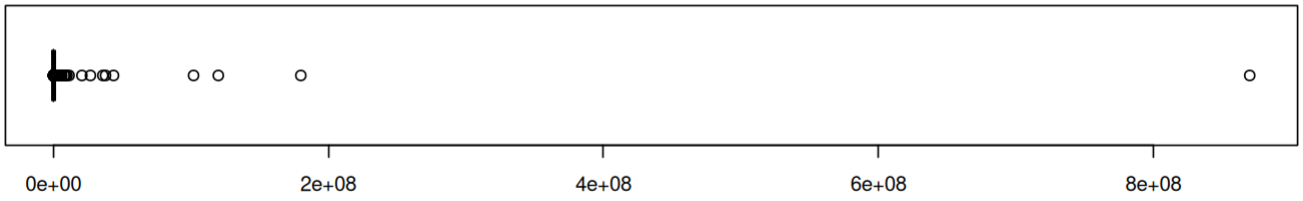
```
sal$salario <- as.numeric(sal$salario)
```

```
summary(sal$salario)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
      0   54000   75000  147570  109884 870000000 
```

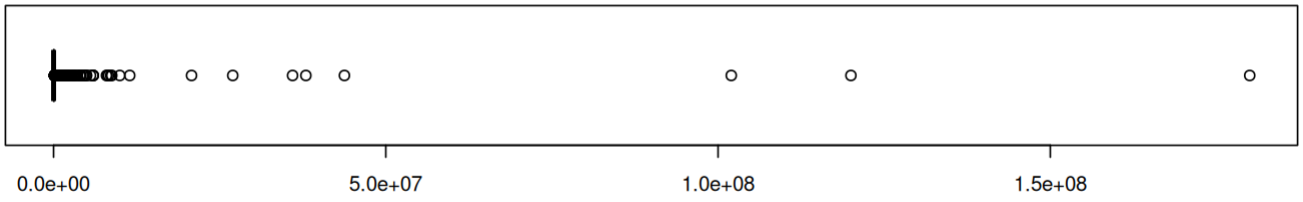
Como estariam distribuídos os salários?

```
dimgraf(12,3)
boxplot(sal$salario, horizontal=TRUE, col="moccasin")
```



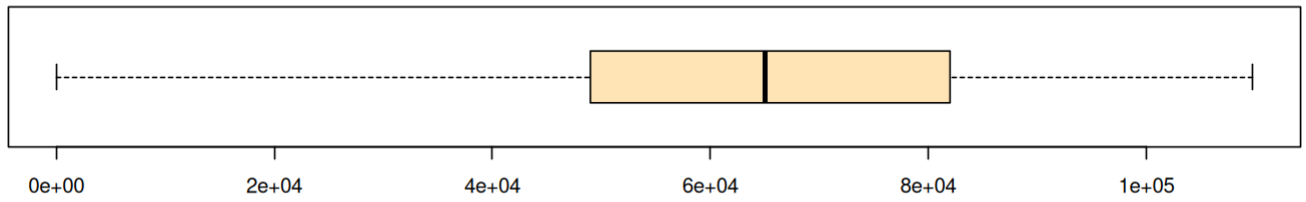
Aparentemente um valor extremo distorceu o boxplot:

```
boxplot(sal$salario[sal$salario < 2e8], horizontal=TRUE, col="moccasin")
```



Ainda há valores extremos, provavelmente provenientes das outras moedas. Como curiosidade, vamos observar até 3º quartil:

```
boxplot(sal$salario[sal$salario < 110000], horizontal=TRUE, col="moccasin")
```



Parece mais razoável. Talvez a exclusão dos valores mais altos, talvez em outras moedas, mostre uma distribuição mais próxima da correta. A real distribuição só será conhecida ao converter as diferentes moedas para uma única, provavelmente a `USD`.

moeda

```
head(sal$moeda)
tail(sal$moeda)
```

1. 'USD'
2. 'GBP'
3. 'USD'
4. 'USD'
5. 'USD'
6. 'USD'

1. 'USD'
2. 'USD'
3. 'USD'
4. 'USD'
5. 'USD'
6. 'USD'

```
unique(sal$moeda)
```

1. 'USD'
2. 'GBP'
3. 'CAD'
4. 'EUR'
5. 'AUD/NZD'
6. 'Other'
7. 'CHF'
8. 'ZAR'
9. 'SEK'
10. 'HKD'
11. 'JPY'

```
sal$moeda <- factor(sal$moeda)
print(summary(sal$moeda))
```

```
AUD/NZD     CAD     CHF     EUR     GBP     HKD     JPY   Other     SEK     USD
    504    1673      37     643    1591       4      23     160      37   23387
    ZAR
     16
```

moeda_outra

```
print(head(sal$moeda_outra,10))
print(tail(sal$moeda_outra,10))
```

```
 [1] "" "" "" "" "" "" "" "" "" ""
 [1] "" "" "" "" "" "" "" "" "" ""
```

```
mo <- sal$moeda_outra
```

Verificando quantas observações não tem valor para esta variável:

```
mo[nchar(mo) == 0] <- NA
(total_na <- sum(is.na(nchar(mo) == 0)))
(total <- length(mo))
(total_na/total)
```

27862

28075

0.992413178984862

Menos que 1% tem valores para esta variável.

As informações adicionais sobre a moeda são muito poucas e confusas. Há pouca chance de agregarem algo à análise.

```
sal$moeda_outra <- NULL
```

pais

```
head(sal$pais,20)
tail(sal$pais,20)
```

1. 'United States'
2. 'United Kingdom'
3. 'US'
4. 'USA'
5. 'US'
6. 'USA'
7. 'USA'
8. 'United States'
9. 'US'
10. 'United States'
11. 'United States'
12. 'USA'
13. 'United States'
14. 'United States'
15. 'Canada'
16. 'United Kingdom'
17. 'USA'
18. 'usa'
19. 'USA'
20. 'United States'

1. 'Canada'

2. ‘Canada’
3. ‘United States’
4. ‘USA’
5. ‘Myanmar’
6. ‘Burma’
7. ‘Nigeria’
8. ‘United States’
9. ‘USA’
10. ‘USA’
11. ‘USA’
12. ‘United States’
13. ‘United States of America’
14. ‘U.S.’
15. ‘USA’
16. ‘United States’
17. ‘USA’
18. ‘China’
19. ‘United States’
20. ‘United States’

```
library(stringr)
p  <- str_to_title(trimws(sal$pais))
```

```
head(unique(p), 30)
```

1. ‘United States’
2. ‘United Kingdom’
3. ‘Us’
4. ‘Usa’
5. ‘Canada’
6. ‘Uk’
7. ‘Scotland’
8. ‘U.s.’
9. ‘The Netherlands’
10. ‘Australia’
11. ‘Spain’
12. ‘England’
13. ‘Finland’
14. ‘United States Of America’
15. ‘France’
16. ‘Germany’
17. ‘Ireland’
18. ‘India’
19. ‘U.s>’
20. ‘Isa’
21. ‘Argentina’
22. ‘Great Britain’
23. ‘United State’
24. ‘U.s.a’
25. ‘Denmark’
26. ‘U.s.a.’
27. ‘America’
28. ‘Netherlands’
29. ‘Switzerland’
30. ‘Bermuda’

```
p[(grepl("u.( )?s(.)?", p, ignore.case=TRUE))]  <- 'U.S.A.'
```

```
p[grepl("states", p, ignore.case = TRUE)]  <- 'U.S.A.'
p[grepl(".nite. sta.e.", p, ignore.case=TRUE)] <- 'U.S.A.'
p[grepl("usa", p, ignore.case=TRUE)] <- 'U.S.A.'
p[grepl("us", p, ignore.case=TRUE)] <- 'U.S.A.'
p[grepl("america", p, ignore.case=TRUE)] <- 'U.S.A.'
p[substring(p,1,8) == "United S"] <- 'U.S.A.'
p[grepl("unitedstates", p, ignore.case=TRUE)] <- 'U.S.A.'
p[grepl("un(i)?(i)?(t|y)(i)?(e|i)(d)? stat(t)?.s",p,ignore.case = TRUE)] <- 'U.S.A.'
p[p == '□□'] <- 'U.S.A.'
p[p == 'ISA'] <- 'U.S.A.'
p[grepl("un(i)?(i)?(t|y)(t)?(e)?(e|i)(d|s)? s(t)?(a)?t.s",p,ignore.case = TRUE)] <- 'U.S.A.'
```

```
p[grepl("united kingdom", p, ignore.case=TRUE)] <- "U.K."
p[grepl("uk$", p, ignore.case=TRUE)] <- 'U.K.'
p[grepl("u.k$", p, ignore.case=TRUE)] <- 'U.K.'
p[grepl("u.k.$", p, ignore.case=TRUE)] <- 'U.K.'
p[grepl("u.k.,", p, ignore.case=TRUE)] <- 'U.K.'
p[grepl("uk ", p, ignore.case=TRUE)] <- 'U.K.'
p[grepl("uk,", p, ignore.case=TRUE)] <- 'U.K.'
p[grepl("unite...in(g)?do*", p, ignore.case=TRUE)] <- "U.K."
p[grepl("great brit.", p, ignore.case = TRUE)] <- "U.K."
p[grepl("englan.", p, ignore.case = TRUE)] <- "U.K."
p[grepl("wales.", p, ignore.case = TRUE)] <- "U.K."
p[grepl("wales", p, ignore.case = TRUE)] <- "U.K."
p[grepl("Scotlan.", p, ignore.case = TRUE)] <- "U.K."
p[grepl("britai.", p, ignore.case = TRUE)] <- "U.K."
p[grepl("london", p, ignore.case = TRUE)] <- "U.K."
```

```
p[grepl("Can.", p, ignore.case = TRUE)] <- 'Canada'
p[p == "Can"] <- 'Canada'
p[p == "Csnada"] <- 'Canada'
```

```
p[grepl("spain", p, ignore.case = TRUE)] <- 'Spain'
p[grepl("brasil", p, ignore.case = TRUE)] <- 'Brazil'
p[grepl("d(a|e)nmark", p, ignore.case=TRUE)] <- 'Denmark'
p[grepl("netherland", p, ignore.case=TRUE)] <- 'The Netherlands'
p[p == "NL"] <- 'The Netherlands'
p[grepl("new zealan.", p, ignore.case = TRUE)] <- "New Zealand"
p[p == "NZ"] <- "New Zealand"
p[grepl("i.dia", p, ignore.case = TRUE)] <- "India"
p[grepl("n(i)?geria", p, ignore.case = TRUE)] <- "Nigeria"
p[grepl("hong kon.", p, ignore.case = TRUE)] <- "Hong Kong"
p[grepl("czechia", p, ignore.case = TRUE)] <- "Czech Republic"
p[substring(p, 1, 5) == 'Italy'] <- 'Italy'
```

```
unique(p)
```

1. 'U.S.A.'
2. 'U.K.'
3. 'Canada'
4. 'The Netherlands'
5. 'Spain'
6. 'Finland'

7. ‘France’
8. ‘Germany’
9. ‘Ireland’
10. ‘India’
11. ‘Isa’
12. ‘Argentina’
13. ‘Denmark’
14. ‘Switzerland’
15. ‘Bermuda’
16. ‘Malaysia’
17. ‘Mexico’
18. ‘South Africa’
19. ‘Belgium’
20. ‘Northern Ireland’
21. ‘Sweden’
22. ‘Hong Kong’
23. ‘Kuwait’
24. ‘Norway’
25. ‘Sri Lanka’
26. ‘Contracts’
27. ‘Greece’
28. ‘Japan’
29. ‘Brazil’
30. ‘Global’
31. ‘Hungary’
32. ‘Luxembourg’
33. ‘Colombia’
34. ‘New Zealand’
35. ‘Trinidad And Tobago’
36. ‘Cayman Islands’
37. ‘U.a.’
38. ‘Czech Republic’
39. ‘Latvia’
40. ‘Puerto Rico’
41. ‘Rwanda’
42. ‘United Arab Emirates’
43. ‘Bangladesh’
44. ‘Romania’
45. ‘Currently Finance’
46. ‘Serbia’
47. ‘Philippines’
48. ‘Poland’
49. ‘Uxz’
50. ‘Turkey’
51. ‘Catalonia’
52. ‘$2,175.84/Year Is Deducted For Benefits’
53. ‘Italy’
54. ‘Jersey, Channel Islands’
55. ‘China’
56. ‘Virginia’
57. ‘Afghanistan’
58. ‘Israel’
59. ‘Hartford’
60. ‘Mainland China’
61. ‘I.s.’
62. ‘Nz’
63. ‘Taiwan’

64. ‘Cambodia’
65. ‘Vietnam’
66. ‘Remote’
67. ‘Singapore’
68. ‘South Korea’
69. ‘Thailand’
70. ‘Lithuania’
71. ‘Eritrea’
72. ‘Indonesia’
73. ‘Is’
74. ‘Cuba’
75. ‘Slovenia’
76. ‘Cote D'ivoire’
77. ‘Somalia’
78. ‘Nl’
79. ‘Nederland’
80. ‘Slovakia’
81. ‘Portugal’
82. ‘Sierra Leone’
83. ‘Uae’
84. ‘International’
85. ‘The Bahamas’
86. ‘I Earn Commission On Sales. If I Meet Quota, I'm Guaranteed Another 16k Min. Last Year I Earned An Additional 27k. It's Not Uncommon For People In My Space To Earn 100k+ After Commission.’
87. ‘Costa Rica’
88. ‘Chile’
89. ‘Qatar’
90. ‘Remote (Philippines)’
91. ‘Nigeria’
92. ‘Panamá’
93. ‘I Was Brought In On This Salary To Help With The Ehr And Very Quickly Was Promoted To Current Position But Compensation Was Not Altered.’
94. ‘Congo’
95. ‘Uruguay’
96. ‘Pakistan’
97. ‘N/A (Remote From Wherever I Want)’
98. ‘Uganda’
99. ‘Malta’
100. ‘Saudi Arabia’
101. ‘Bulgaria’
102. ‘Estonia’
103. ‘Morocco’
104. ‘Africa’
105. ‘Ecuador’
106. ‘Zimbabwe’
107. ‘Ghana’
108. ‘San Francisco’
109. ‘Luxemburg’
110. ‘Ua’
111. ‘Croatia’
112. ‘Ukraine’
113. ‘Y’
114. ‘United Y’
115. ‘Isle Of Man’
116. ‘Europe’
117. ‘California’
118. ‘México’

119. ‘Jamaica’
120. ‘Kenya’
121. ‘Jordan’
122. ‘Argentina But My Org Is In Thailand’
123. ‘Na’
124. ‘Policy’
125. ‘Liechtenstein’
126. ‘Company In Germany. I Work From Pakistan.’
127. ‘Bosnia And Herzegovina’
128. ‘Ss’
129. ‘Dbfemf’
130. ‘Loutreland’
131. ‘Ff’
132. ‘Myanmar’
133. ‘Burma’

```
pfat <- factor(p)
summary(pfat)
```

U.S.A. 23606Canada
1686U.K.
1593Germany
195New Zealand
126Ireland
124The Netherlands
88France
68Spain
49Sweden
41Switzerland
38Belgium
35Japan
29Denmark
24India
22Singapore
19South Africa
19Finland
16Italy
15Israel
14Norway
14Malaysia
13Brazil
12Philippines
10Poland
10China
9Mexico
7Czech Republic
6Nz
6Thailand
6Argentina
5Colombia
5Greece
5Hong Kong
5Nigeria
5Pakistan
5Portugal
5Romania
5South Korea

5Latvia
4Puerto Rico
4Taiwan
4Chile
3Ghana
3Isa
3Kenya
3Northern Ireland
3Bangladesh
2Bermuda
2Bulgaria
2Croatia
2Estonia
2Hungary
2I.s.
2Indonesia
2Lithuania
2Luxembourg
2Remote
2Saudi Arabia
2Slovenia
2Sri Lanka
2Turkey
2Vietnam
2Zimbabwe
2$2,175.84/Year Is Deducted For Benefits
1Afghanistan
1Africa
1Argentina But My Org Is In Thailand
1Bosnia And Herzegovina
1Burma
1California
1Cambodia
1Catalonia
1Cayman Islands
1Company In Germany. I Work From Pakistan.
1Congo
1Contracts
1Costa Rica
1Cote D'ivoire
1Cuba
1Currently Finance
1Dbfemf
1Ecuador
1Eritrea
1Europe
1Ff
1Global
1Hartford
1I Earn Commission On Sales. If I Meet Quota, I'm Guaranteed Another 16k Min. Last Year I Earned An Additional 27k. It's Not Uncommon For People In My Space To Earn 100k+ After Commission.
1I Was Brought In On This Salary To Help With The Ehr And Very Quickly Was Promoted To Current Position But Compensation Was Not Altered.
1International
1Is
1Isle Of Man
1Jamaica

1Jersey, Channel Islands
1Jordan
1Kuwait
1Liechtenstein
1Loutreland
1(Other)
34

```
total <- nrow(sal)
topo  <- 23606+1686+1592+195+126+124+88+68+49+41+38+35+29+24+22+19+16+15+14+13+12+10+10
topo / total * 100
```

99.1344612644702

Compondo até a categoria ter 99% com países conhecidos, temos:

```
p[ p != 'U.S.A.' &   p != 'Canada' &   p != 'U.K.' &   p != 'Germany' &
   p != 'New Zealand' &   p != 'Ireland' &   p != 'France' &
   p != 'Netherlands' &   p != 'Spain' &   p != 'Sweden' &  p != 'Switzerland' &
   p != 'Belgium' &  p != 'Japan' &  p != 'Denmark' &  p != 'India' &
   p != 'Singapore' &  p != 'South Africa' &  p != 'Finland' &  p != 'Italy' &
   p != 'Israel' &  p != 'Norway' &  p != 'Malaysia' &  p != 'Brazil' &
   p != 'Philippines' &  p != 'Poland' ] <- 'other'
p <- factor(p)
```

```
sal$pais <- p
```

```
summary(sal)
```

```
     tempo                          idade
 Min.   :2021-04-27 11:02:10.00   -18  :   14
 1st Qu.:2021-04-27 14:21:35.50   18-24: 1224
 Median :2021-04-28 13:57:11.00   25-34:12660
 Mean   :2021-05-15 10:18:53.21   35-44: 9902
 3rd Qu.:2021-04-29 20:31:53.50   45-54: 3189
 Max.   :2024-09-17 12:03:31.00   55-64:  992
                                  65+  :   94
                                industria       cargo
 other                               :5465   Length:28075
 computing or tech                   :4704   Class :character
 education (higher education)        :2466   Mode  :character
 nonprofits                          :2419
 health care                         :1896
 government and public administration:1890
 (Other)                             :9235
    salario              bonus               moeda       salario_contexto
 Min.   :        0   Min.   :        0   USD    :23387   Length:28075
 1st Qu.:    54000   1st Qu.:        0   CAD    : 1673   Class :character
 Median :    75000   Median :     2000   GBP    : 1591   Mode  :character
 Mean   :   147570   Mean   :    18159   EUR    :  643
 3rd Qu.:   109884   3rd Qu.:    10000   AUD/NZD:  504
 Max.   :870000000   Max.   :120000000   Other  :  160
                     NA's   :7302        (Other):  117
          pais        estado_eua           cidade          anos_exp_total
 U.S.A.      :23606   Length:28075       Length:28075       Length:28075
 Canada      : 1686   Class :character   Class :character   Class :character
```

```
U.K.        : 1593   Mode  :character   Mode  :character   Mode  :character
other       :  297
Germany     :  195
New Zealand:  126
(Other)     :  572
anos_exp_area        educacao             sexo              etnia
Length:28075       Length:28075       Length:28075       Length:28075
Class :character   Class :character   Class :character   Class :character
Mode  :character   Mode  :character   Mode  :character   Mode  :character
```

estado_eua

```
head(sal$estado_eua)
tail(sal$estado_eua)
```

1. 'Massachusetts'
2. ''
3. 'Tennessee'
4. 'Wisconsin'
5. 'South Carolina'
6. 'New Hampshire'

1. 'Nevada'
2. 'Georgia'
3. 'Georgia'
4. ''
5. 'Georgia'
6. 'Georgia'

```
unique(sal$estado_eua)
```

1. 'Massachusetts'
2. ''
3. 'Tennessee'
4. 'Wisconsin'
5. 'South Carolina'
6. 'New Hampshire'
7. 'Arizona'
8. 'Missouri'
9. 'Florida'
10. 'Pennsylvania'
11. 'Michigan'
12. 'Minnesota'
13. 'Illinois'
14. 'California'
15. 'Georgia'
16. 'Ohio'
17. 'District of Columbia'
18. 'Maryland'
19. 'Texas'
20. 'Virginia'
21. 'North Carolina'

22. ‘New York’
23. ‘New Jersey’
24. ‘Rhode Island’
25. ‘Colorado’
26. ‘Oregon’
27. ‘Washington’
28. ‘Indiana’
29. ‘Iowa’
30. ‘Nebraska’
31. ‘Oklahoma’
32. ‘Maine’
33. ‘Connecticut’
34. ‘South Dakota’
35. ‘West Virginia’
36. ‘Idaho’
37. ‘Louisiana’
38. ‘Montana’
39. ‘Kentucky’
40. ‘North Dakota’
41. ‘Kansas’
42. ‘Vermont’
43. ‘Arkansas’
44. ‘Alabama’
45. ‘Nevada’
46. ‘Delaware’
47. ‘New Mexico’
48. ‘Hawaii’
49. ‘Utah’
50. ‘Mississippi’
51. ‘Kentucky, Ohio’
52. ‘District of Columbia, Virginia’
53. ‘District of Columbia, Maryland’
54. ‘Alaska’
55. ‘Arizona, Washington’
56. ‘Georgia, New York’
57. ‘California, Colorado’
58. ‘California, Oregon’
59. ‘District of Columbia, Maryland, Pennsylvania, Virginia’
60. ‘Arizona, California’
61. ‘North Carolina, Utah’
62. ‘Wyoming’
63. ‘Ohio, Wyoming’
64. ‘Georgia, Tennessee’
65. ‘Massachusetts, Oregon’
66. ‘Alabama, Montana’
67. ‘Alabama, District of Columbia’
68. ‘California, Pennsylvania’
69. ‘New Jersey, Pennsylvania’
70. ‘Georgia, Washington’
71. ‘Alaska, Maryland’
72. ‘Michigan, South Carolina’
73. ‘Massachusetts, Rhode Island’
74. ‘Georgia, Minnesota’
75. ‘Colorado, Nevada’
76. ‘Maine, Massachusetts, New Hampshire, North Carolina’
77. ‘Alabama, Minnesota, Nevada’
78. ‘New Jersey, New York’

79. ‘Arizona, Utah’
80. ‘Alabama, Kansas’
81. ‘California, Oklahoma’
82. ‘Illinois, Wisconsin’
83. ‘Illinois, Kentucky’
84. ‘Arizona, California, Nevada, Texas’
85. ‘Alaska, Idaho, Oregon, Utah, Washington’
86. ‘Massachusetts, Pennsylvania’
87. ‘Nevada, Oregon’
88. ‘New Jersey, Virginia’
89. ‘Montana, Wyoming’
90. ‘Colorado, Massachusetts’
91. ‘District of Columbia, Maryland, Virginia’
92. ‘Massachusetts, Vermont’
93. ‘Massachusetts, New Hampshire’
94. ‘Arkansas, Iowa, Massachusetts, Ohio, Wyoming’
95. ‘New York, Texas’
96. ‘California, Montana’
97. ‘Iowa, Utah, Vermont’
98. ‘Texas, Virginia’
99. ‘Utah, Vermont’
100. ‘Arkansas, Illinois’
101. ‘Georgia, Massachusetts’
102. ‘Maryland, Virginia’
103. ‘Florida, Georgia, South Carolina’
104. ‘Arkansas, Idaho, Kansas, Louisiana, Michigan, Mississippi, Nevada, New York, South Carolina, Tennessee, Washington’
105. ‘California, Texas’
106. ‘Indiana, Ohio’
107. ‘Ohio, Washington’
108. ‘Kansas, Missouri’
109. ‘Colorado, Illinois’
110. ‘Arizona, Hawaii, Illinois, Michigan, Utah, Wyoming’
111. ‘California, New Jersey’
112. ‘Louisiana, Washington’
113. ‘Maryland, New York’
114. ‘District of Columbia, Washington’
115. ‘Delaware, Pennsylvania’
116. ‘Illinois, North Carolina’
117. ‘Indiana, Massachusetts’
118. ‘Florida, New Hampshire, Wisconsin’
119. ‘Pennsylvania, Rhode Island’
120. ‘New York, Oregon, Vermont’
121. ‘Iowa, Nebraska’
122. ‘California, New York’
123. ‘Arizona, New York’
124. ‘California, District of Columbia, Illinois, Iowa, Maryland, Minnesota’
125. ‘Oregon, Washington’
126. ‘New York, Virginia’
127. ‘Mississippi, Missouri’
128. ‘California, Maryland’
129. ‘California, Illinois, Massachusetts, North Carolina, South Carolina, Virginia’
130. ‘Alabama, California’
131. ‘Michigan, Texas, Washington’
132. ‘Alabama, Oregon’
133. ‘Alabama, Alaska, Arizona’
134. ‘Alabama, South Carolina’

135. ‘Colorado, Delaware, New Jersey, West Virginia, Wyoming’

Aparentemente como o estado também devia ser uma escolha múltipla, há respondentes com múltiplos estados; alguns casos com mais que 10 estados. Descartando.

```
sal$estado_eua <- NULL
```

cidade

```
head(sal$cidade)
tail(sal$cidade)
```

1. ‘Boston’
2. ‘Cambridge’
3. ‘Chattanooga’
4. ‘Milwaukee’
5. ‘Greenville’
6. ‘Hanover’

1. ‘Las Vegas’
2. ‘Milledgeville’
3. ‘Johns Creek’
4. ‘Shenzhen’
5. ‘Tucker’
6. ‘Roswell’

```
unique(sal$cidade)
```

1. ‘Boston’
2. ‘Cambridge’
3. ‘Chattanooga’
4. ‘Milwaukee’
5. ‘Greenville’
6. ‘Hanover’
7. ‘Columbia’
8. ‘Yuma’
9. ‘St. Louis’
10. ‘Palm Coast’
11. ‘Boston, MA’
12. ‘Scranton’
13. ‘Detroit’
14. ‘Saint Paul’
15. ‘Remote’
16. ‘Lincoln’
17. ‘Chicago’
18. ‘Pomona’
19. ‘Atlanta’
20. ‘Boca Raton’
21. ‘Philadelphia’
22. ‘Toronto’
23. ‘Dayton’
24. ‘Bradenton’
25. ‘Ann Arbor’
26. ‘Washington DC’
27. ‘Silver Spring’
28. ‘Washington’

29. ‘San Antonio’
30. ‘Minneapolis’
31. ‘Washington, DC’
32. ‘Richmond’
33. ‘Research Triangle’
34. ‘Kalamazoo’
35. ‘Manhattan’
36. ‘Sacramento’
37. ‘Dallas’
38. ‘waynesboro’
39. ‘Liverpool’
40. ‘Pittsburgh’
41. ‘Arlington, VA’
42. ‘Chapel Hill’
43. ‘Vancouver’
44. ‘Berkeley’
45. ‘Apple Valley’
46. ‘Troy’
47. ‘DC’
48. ‘Linden’
49. ‘Bristol’
50. ‘Champaign’
51. ‘Nova Scotia’
52. ‘Ottawa’
53. ‘Bryan’
54. ‘Providence’
55. ‘District of Columbia’
56. ‘Bloomington’
57. ‘Denver’
58. ‘New York City’
59. ‘Colorado Springs’
60. ‘Southampton’
61. ‘prefer not to answer’
62. ‘NYC (remotely)’
63. ‘Raleigh’
64. ‘New York’
65. ‘—’
66. ‘Edinburgh’
67. ‘St. Paul’
68. ‘Philadelphia (suburbs)’
69. ‘Kitchener, Ontario’
70. ‘Portland’
71. ‘Nashville’
72. ‘Newcastle, UK’
73. ‘Seattle’
74. ‘Indianapolis’
75. ‘Baltimore’
76. ‘Solon’
77. ‘Montreal’
78. ‘Gainesville’
79. ‘Lafayette’
80. ‘Phoenix’
81. ‘Boulder’
82. ‘Austin’
83. ‘Halifax’
84. ‘Portsmouth’
85. ‘Nottingham’

86. ‘Prefer not to answer’
87. ‘Glasgow’
88. ‘The hague’
89. ‘Calgary’
90. ‘Kansas City’
91. ‘Dubuque’
92. ‘London, ON CAN’
93. ‘Cardiff’
94. ‘San Diego’
95. ‘Belmont’
96. ‘Philadelphia’
97. ‘Whippany’
98. ‘Jefferson’
99. ‘Manassas’
100. ‘Detroit Metro area’
101. ‘Perth’
102. ‘Newtown Square’
103. ‘Small city, remote, national company’
104. ‘Memphis’
105. ‘Grand Rapids’
106. ‘London’
107. ‘N/A’
108. ‘New York’
109. ‘Omaha’
110. ‘Oxford’
111. ‘McLean’
112. ‘Ardmore’
113. ‘Columbus’
114. ‘Shaker Hts’
115. ‘Charleston’
116. ‘helsinki’
117. ‘Madison’
118. ‘Mishawaka’
119. ‘Tulsa’
120. ‘Tampa’
121. ‘Lambertville’
122. ‘Chatham’
123. ‘Skokie’
124. ‘London’
125. ‘Alpharetta’
126. ‘Northeast Florida’
127. ‘New York City’
128. ‘Altamonte Springs’
129. ‘Brighton’
130. ‘San Francisco’
131. ‘Portland’
132. ‘El Paso’
133. ‘Research Triangle region’
134. ‘Lyon’
135. ‘Merced’
136. ‘Aberdeen’
137. ‘Birmingham’
138. ‘Gorham/Portland’
139. ‘Westchester/NYC’
140. ‘San Jose’
141. ‘Grandville’
142. ‘Remote Worker’

143. ‘Houston’
144. ‘Eau Claire’
145. ‘buffalo’
146. ‘West Lafayette’
147. ‘Too identifiable’
148. ‘Braintree’
149. ‘Clinton’
150. ‘Woburn’
151. ‘Boston’
152. ‘Anaheim’
153. ‘Orlando’
154. ‘Appleton’
155. ‘Jacksonville’
156. ‘Durham’
157. ‘Stamford’
158. ‘Rural Ontario’
159. ‘Newport’
160. ‘Los Angeles’
161. ‘New Brunswick’
162. ‘Chicago area mostly, but also the US and Canada’
163. ‘Spokane’
164. ‘Greater Boston area’
165. ‘Chicago (remote)’
166. ‘Rapid City’
167. ‘Mount Prospect’
168. ‘Huntington (remote, HQ is in Charleston)’
169. ‘Düsseldorf’
170. ‘Charlotte’
171. ‘Flagstaff’
172. ‘Carlinville’
173. ‘Chaska’
174. ‘Omaha, NE’
175. ‘Western Mass’
176. ‘Joliet’
177. ‘Shingle Springs’
178. ‘Plant City’
179. ‘NYC’
180. ‘Lisle’
181. ‘Salem’
182. ‘Platteville’
183. ‘Remote (live in Nottingham)’
184. ‘Suitland’
185. ‘Rockville’
186. ‘Winnipeg’
187. ‘Indianapolis’
188. ‘Dublin’
189. ‘North East’
190. ‘Baltimore’
191. ‘Houston Area’
192. ‘Pittsburgh’
193. ‘Charlottesville’
194. ‘Idaho Falls’
195. ‘ROCHESTER’
196. ‘Metro Boston’
197. ‘Twin cities metro area’
198. ‘Duluth’
199. ‘South Hadley’

200. ‘DFW area’
201.
202. ‘Chicago/Evanston’
203. ‘Nola’
204. ‘Charlton’
205. ‘Hambrug’
206. ‘Dallas Area’
207. ‘Napier’
208. ‘wellington’
209. ‘Upper Hutt’
210. ‘Jefferson city’
211. ‘Lennox’
212. ‘Wales’
213. ‘Blooming Prairie’
214. ‘All, travel’
215. ‘Gelderland’
216. ‘Lawrence’
217. ‘Freemont’
218. ‘Larnaca’
219. ‘Stl’
220. ‘SMYRNA’
221. ‘Milwaukee (remote)’
222. ‘Horst’
223. ‘wilkes barre’
224. ‘Coeur d Alene’
225. ‘Harmans’
226. ‘Laguna hills’
227. ‘hong kong’
228. ‘London/Remote’
229. ‘middleburg heights’
230. ‘Romeo’
231. ‘Thetford Center’
232. ‘Cumberland’
233. ‘Dana Point’
234. ‘La Crosse’
235. ‘Louisville, CO’
236. ‘Wirral’
237. ‘Marlborough’
238. ‘Ceres’
239. ‘Bountiful’
240. ‘NEW YORK’
241. ‘saddlebrook’
242. ‘Stowe’
243. ‘Schaan’
244. ‘Pasific city’
245. ‘Goleta’
246. ‘Layton’
247. ‘Skowhegan’
248. ‘Lahore’
249. ‘Sorry, no’
250. ‘Corner Brook’
251. ‘Clearwater’
252. ‘Williston’
253. ‘NoVA’
254. ‘Alabama’
255. ‘Gardner’
256. ‘Friedrichsdorf’

257. ‘Pontiac’
258. ‘Panther Junction’
259. ‘pula’
260. ‘Highgate’
261. ‘Winona’
262. ‘Fargo, ND’
263. ‘Addis’
264. ‘Solothurn’
265. ‘Antwerpen’
266. ‘Westlake Village’
267. ‘Huntington Beach’
268. ‘North Chicago’
269. ‘Benton harbor’
270. ‘Donegal’
271. ‘Española’
272. ‘SF Bay Area (Remote)’
273. ‘Shawnee’
274. ‘West Metro’
275. ‘Ionia’
276. ‘Idaho’
277. ‘Pine Island’
278. ‘Kennedy Space Center’
279. ‘Sterling Heights’
280. ‘Adairsville’
281. ‘santa barbara’
282. ‘North canton’
283. ‘albany’
284. ‘Houghton’
285. ‘South Portland’
286. ‘Hershey’
287. ‘Brockton’
288. ‘Houston TX’
289. ‘Edina’
290. ‘Paterson’
291. ‘Safford’
292. ‘Mahomet’
293. ‘Betavia’
294. ‘Louisburg’
295. ‘Branchburg’
296. ‘Loughborough’
297. ‘Mt. Vernon’
298. ‘L'Islet’
299. ‘Milwaukee/Remote’
300. ‘Poznań’
301. ‘south portland’
302. ‘British columbia’
303. ‘Sherwood Park’
304. ‘Kincardine’
305. ‘Xxxx’
306. ‘Latrobe’
307. ‘Orange County’
308. ‘durham’
309. ‘Powhatan, VA’
310. ‘mannheim’
311. ‘Bolivar’
312. ‘Swift current’
313. ‘Palatine’

314. ‘Oconomowoc’
315. ‘Salzburg’
316. ‘RIchmond (remote company though)’
317. ‘New Bedford’
318. ‘Raleigh Durham area’
319. ‘BANGALORE’
320. ‘Corona Del Mar’
321. ‘Secaucus, NJ’
322. ‘dino’
323. ‘Peterborough’
324. ‘New York City (remote)’
325. ‘Medellin’
326. ‘Westford’
327. ‘Port orchard’
328. ‘Bemidji’
329. ‘Aurora’
330. ‘Fort worth’
331. ‘Nicosia’
332. ‘Chicago / FT Remote’
333. ‘Seattle (remote)’
334. ‘castellon’
335. ‘Banja Luka’
336. ‘Vine Grove’
337. ‘Romeoville’
338. ‘St Pete’
339. ‘Falls Churcv’
340. ‘Darwin’
341. ‘Amersfoort’
342. ‘SA’
343. ‘Neptune, NJ’
344. ‘Louisville, KY’
345. ‘De Soto’
346. ‘Bengaluru’
347. ‘Twentynine Palms’
348. ‘NYC Suburb’
349. ‘NW’
350. ‘All over’
351. ‘Victorua’
352. ‘Irvine, CA’
353. ‘Singaporw’
354. ‘Alma’
355. ‘Orlando Metro Area’
356. ‘Gdańsk’
357. ‘PORT HARCOURT’
358. ‘Galena’
359. ‘Newbury’
360. ‘Colorado’
361. ‘Istanbul’
362. ‘Honolulu’
363. ‘Danvers, MA (North of Boston)’
364. ‘Jersey CIty’
365. ‘bristol’
366. ‘århus’
367. ‘Spotsylvania’
368. ‘Diamond bar’
369. ‘Maitland, NSW’
370. ‘Banglore’

371. ‘northampton’
372. ‘Palma de mallorca’
373. ‘karachi’
374. ‘St.Louis’
375. ‘East lansing’
376. ‘denton’
377. ‘irving’
378. ‘s’
379. ‘dhgbfv’
380. ‘Bogotá’
381. ‘Al Khobar’
382. ‘i am a student’
383. ‘pubw’
384. ‘Jos’
385. ‘Rice’
386. ‘Overland park’
387. ‘N/A - Remote’
388. ‘Stgo’
389. ‘LOUTRECITY’
390. ‘birmingham’
391. ‘Daleville’
392. ‘Newyork’
393. ‘test’
394. ‘ff’
395. ‘Alabama’
396. ‘Boise-but remote. Company located in Boston’
397. ‘Kochi’
398. ‘Wentzville’
399. ‘Yangon’
400. ‘Macon’
401. ‘Shenzhen’

Seria interessante manter como curiosidade, porém é de pouca utilidade em uma análise.

```
sal$cidade <- NULL
```

anos_exp_total

```
unique(sal$anos_exp_total)
```

1. ‘5-7 years’
2. ‘8 - 10 years’
3. ‘2 - 4 years’
4. ‘21 - 30 years’
5. ‘11 - 20 years’
6. ‘1 year or less’
7. ‘41 years or more’
8. ‘31 - 40 years’

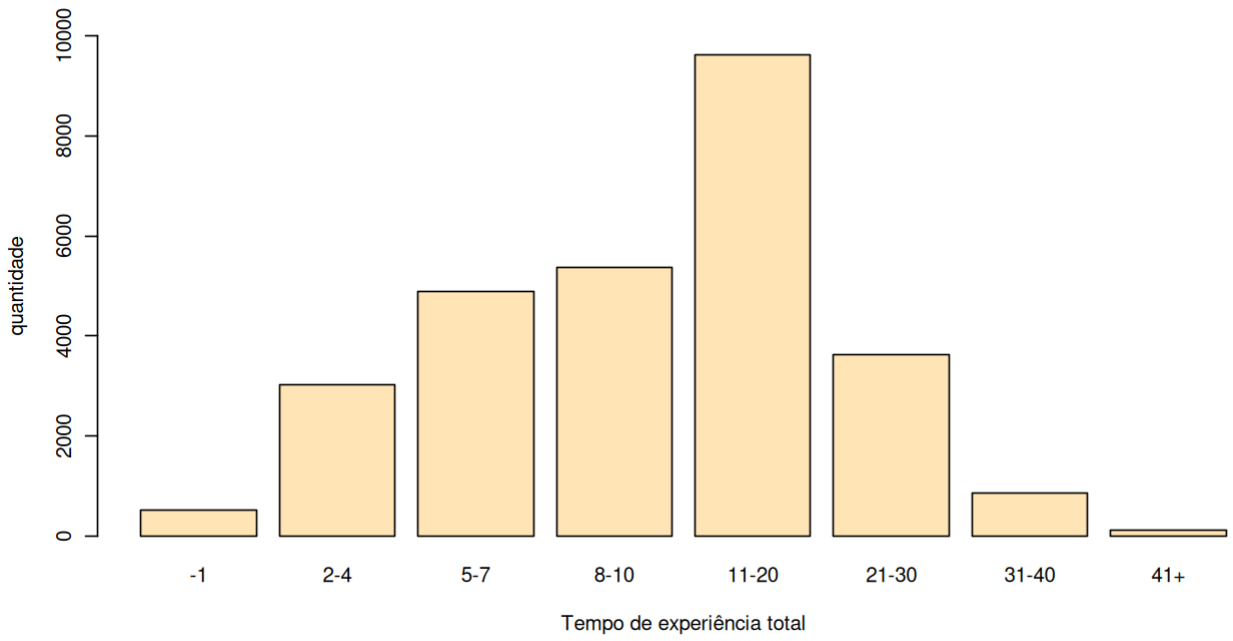
```
aet <- sal$anos_exp_total
aet[aet == '1 year or less'   ] <-    "-1"
aet[aet == '2 - 4 years'      ] <-   "2-4"
aet[aet == '5-7 years'        ] <-   "5-7"
aet[aet == '8 - 10 years'     ] <-  "8-10"
aet[aet == '11 - 20 years'    ] <- "11-20"
aet[aet == '21 - 30 years'    ] <- "21-30"
```

```r
aet[aet == '31 - 40 years'    ] <- "31-40"
aet[aet == '41 years or more' ] <-   "41+"
ordem <- c("-1", "2-4", "5-7", "8-10", "11-20",
           "21-30", "31-40", "41+")
aet <- factor(aet, ordered = TRUE, levels = ordem)
```

```r
print(summary(aet))
```

```
   -1   2-4   5-7  8-10 11-20 21-30 31-40   41+
  527  3030  4883  5379  9625  3637   870   124
```

```r
dimgraf(11,6)
barplot(table(aet),ylim=c(0,10000), col="moccasin",
        xlab="Tempo de experiência total", ylab="quantidade")
```




```r
sal$anos_exp_total <- aet
```

```r
summary(sal)
```

```
     tempo                         idade
 Min.   :2021-04-27 11:02:10.00   -18  :   14
 1st Qu.:2021-04-27 14:21:35.50   18-24: 1224
 Median :2021-04-28 13:57:11.00   25-34:12660
 Mean   :2021-05-15 10:18:53.21   35-44: 9902
 3rd Qu.:2021-04-29 20:31:53.50   45-54: 3189
 Max.   :2024-09-17 12:03:31.00   55-64:  992
                                  65+  :   94
                               industria       cargo
 other                              :5465   Length:28075
 computing or tech                  :4704   Class :character
 education (higher education)       :2466   Mode  :character
 nonprofits                         :2419
 health care                        :1896
```

```
 government and public administration:1890
 (Other)                              :9235
    salario              bonus                moeda       salario_contexto
 Min.   :        0   Min.   :        0   USD    :23387   Length:28075
 1st Qu.:    54000   1st Qu.:        0   CAD    : 1673   Class :character
 Median :    75000   Median :     2000   GBP    : 1591   Mode  :character
 Mean   :   147570   Mean   :    18159   EUR    :  643
 3rd Qu.:   109884   3rd Qu.:    10000   AUD/NZD:  504
 Max.   :870000000   Max.   :120000000   Other  :  160
                     NA's   :7302        (Other):  117
          pais       anos_exp_total anos_exp_area        educacao
 U.S.A.     :23606   11-20  :9625   Length:28075       Length:28075
 Canada     : 1686   8-10   :5379   Class :character   Class :character
 U.K.       : 1593   5-7    :4883   Mode  :character   Mode  :character
 other      :  297   21-30  :3637
 Germany    :  195   2-4    :3030
 New Zealand:  126   31-40  : 870
 (Other)    :  572   (Other): 651
     sexo               etnia
 Length:28075       Length:28075
 Class :character   Class :character
 Mode  :character   Mode  :character
```

anos_exp_area

```
unique(sal$anos_exp_area)
```

1. '5-7 years'
2. '2 - 4 years'
3. '21 - 30 years'
4. '11 - 20 years'
5. '1 year or less'
6. '8 - 10 years'
7. '31 - 40 years'
8. '41 years or more'

```
ae <- sal$anos_exp_area
ae[ae == '1 year or less'   ] <-    "-1"
ae[ae == '2 - 4 years'      ] <-   "2-4"
ae[ae == '5-7 years'        ] <-   "5-7"
ae[ae == '8 - 10 years'     ] <-  "8-10"
ae[ae == '11 - 20 years'    ] <- "11-20"
ae[ae == '21 - 30 years'    ] <- "21-30"
ae[ae == '31 - 40 years'    ] <- "31-40"
ae[ae == '41 years or more' ] <-   "41+"
ordem <- c("-1", "2-4", "5-7", "8-10", "11-20",
           "21-30", "31-40", "41+")
ae <- factor(ae, ordered = TRUE, levels = ordem)
```

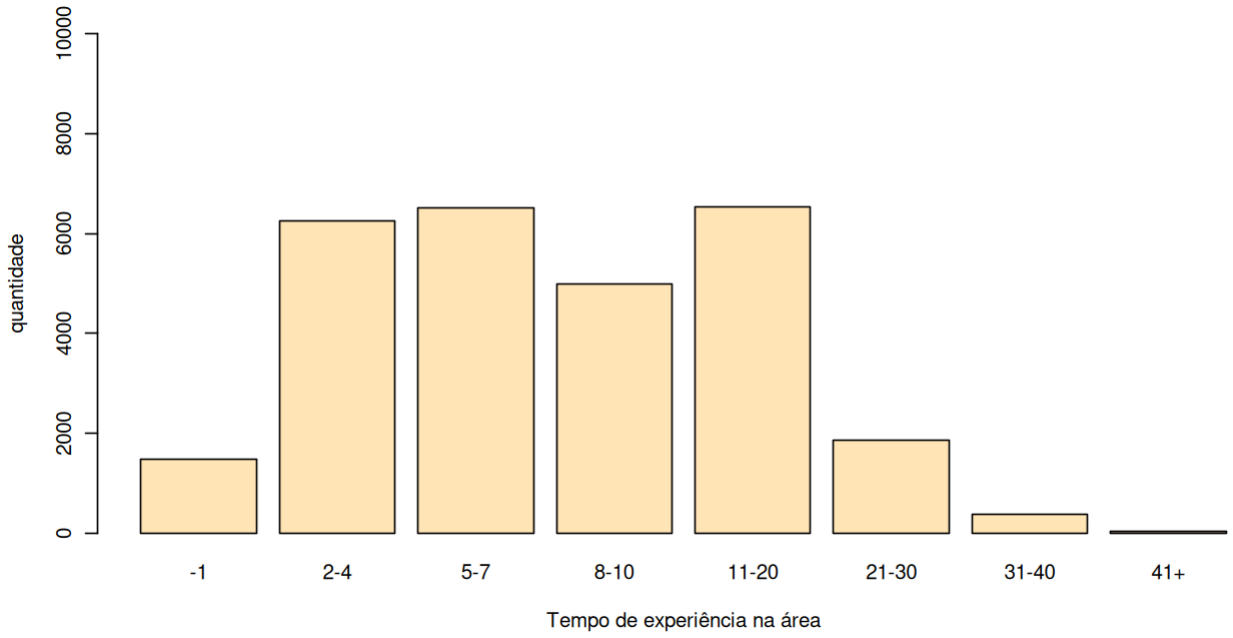
```
print(summary(aet))
```

```
   -1   2-4   5-7  8-10 11-20 21-30 31-40   41+
  527  3030  4883  5379  9625  3637   870   124
```

```
print(summary(ae))
```

```
   -1   2-4   5-7  8-10 11-20 21-30 31-40   41+
 1492  6252  6520  4982  6537  1868   383    41
```

```
dimgraf(11,6)
barplot(table(ae),ylim=c(0,10000), col="moccasin",
        xlab="Tempo de experiência na área", ylab="quantidade")
```




```
sal$anos_exp_area <- ae
```

```
summary(sal)
```

```
    tempo                         idade
Min.   :2021-04-27 11:02:10.00   -18  :   14
1st Qu.:2021-04-27 14:21:35.50   18-24: 1224
Median :2021-04-28 13:57:11.00   25-34:12660
Mean   :2021-05-15 10:18:53.21   35-44: 9902
3rd Qu.:2021-04-29 20:31:53.50   45-54: 3189
Max.   :2024-09-17 12:03:31.00   55-64:  992
                                 65+  :   94
                              industria       cargo
other                              :5465   Length:28075
computing or tech                  :4704   Class :character
education (higher education)       :2466   Mode  :character
nonprofits                         :2419
health care                        :1896
government and public administration:1890
(Other)                            :9235
   salario              bonus               moeda       salario_contexto
```

```
 Min.   :        0   Min.   :        0   USD    :23387   Length:28075
 1st Qu.:    54000   1st Qu.:        0   CAD    : 1673   Class :character
 Median :    75000   Median :     2000   GBP    : 1591   Mode  :character
 Mean   :   147570   Mean   :    18159   EUR    :  643
 3rd Qu.:   109884   3rd Qu.:    10000   AUD/NZD:  504
 Max.   :870000000   Max.   :120000000   Other  :  160
                     NA's   :7302        (Other):  117
          pais       anos_exp_total anos_exp_area    educacao
 U.S.A.     :23606   11-20  :9625   11-20  :6537   Length:28075
 Canada     : 1686   8-10   :5379   5-7    :6520   Class :character
 U.K.       : 1593   5-7    :4883   2-4    :6252   Mode  :character
 other      :  297   21-30  :3637   8-10   :4982
 Germany    :  195   2-4    :3030   21-30  :1868
 New Zealand:  126   31-40  : 870   -1     :1492
 (Other)    :  572   (Other): 651   (Other): 424
     sexo              etnia
 Length:28075       Length:28075
 Class :character   Class :character
 Mode  :character   Mode  :character
```

educacao

```
ed <- sal$educacao
```

```
head(ed)
tail(ed)
```

1. 'Master's degree'
2. 'College degree'
3. 'College degree'
4. 'College degree'
5. 'College degree'
6. 'Master's degree'

1. 'Master's degree'
2. 'Master's degree'
3. 'Master's degree'
4. 'Master's degree'
5. 'College degree'
6. 'Some college'

```
unique(ed)
summary(factor(ed))
```

1. 'Master's degree'
2. 'College degree'
3. 'PhD'
4. ''
5. 'Some college'
6. 'High School'
7. 'Professional degree (MD, JD, etc.)'

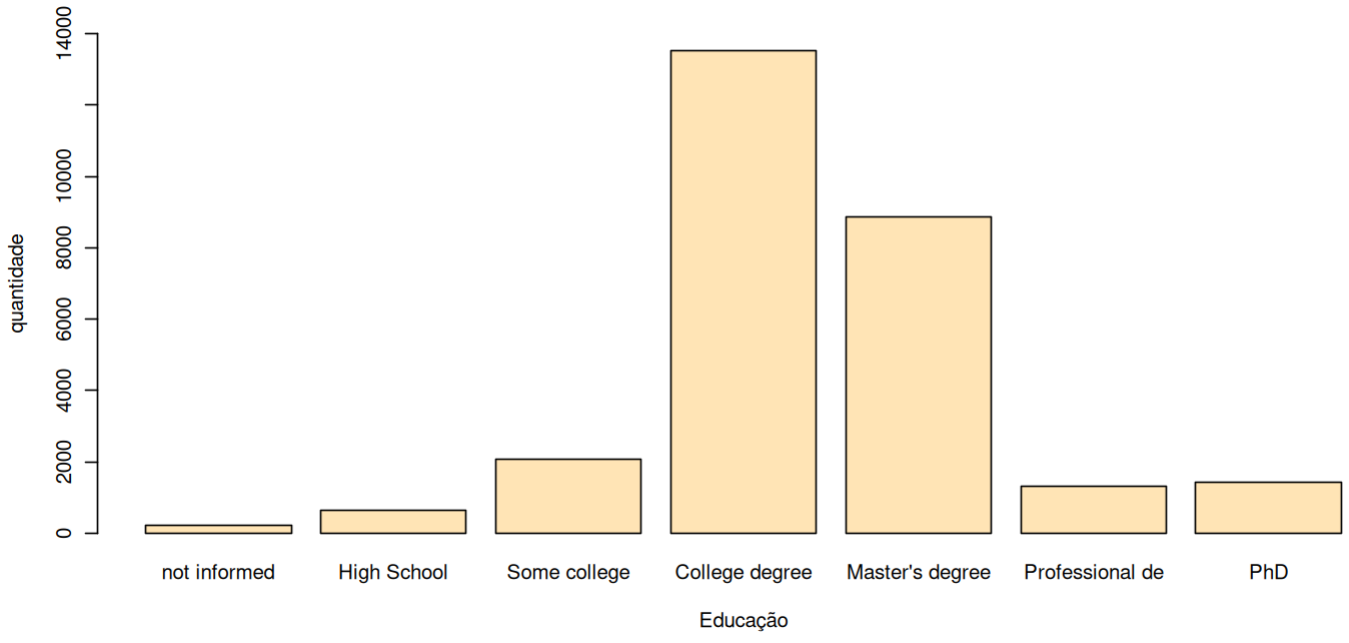
1 222College degree
13527High School
639Master's degree
8869PhD
1426Professional degree (MD, JD, etc.)
1324Some college
2068

```
ed[ed == ""] <- "not informed"
ordem <- c("not informed","High School","Some college",
           "College degree","Master's degree",
           "Professional degree (MD, JD, etc.)", "PhD")
ed2 <- substring(ed, 1, 15)
ed  <- factor(ed,  ordered = TRUE, levels = ordem)
ed2 <- factor(ed2, ordered = TRUE, levels = substring(ordem,1,15))
```

```
print(summary(ed))
```

```
                   not informed                        High School
                            222                                639
                   Some college                      College degree
                           2068                              13527
             Master's degree    Professional degree (MD, JD, etc.)
                           8869                               1324
                            PhD
                           1426
```

```
dimgraf(12,6)
barplot(table(ed2),ylim=c(0,14000), col="moccasin",
        xlab="Educação", ylab="quantidade")
```




```
sal$educacao <- ed
```

```
summary(sal)
```

```
     tempo                          idade
 Min.   :2021-04-27 11:02:10.00   -18  :   14
 1st Qu.:2021-04-27 14:21:35.50   18-24: 1224
 Median :2021-04-28 13:57:11.00   25-34:12660
 Mean   :2021-05-15 10:18:53.21   35-44: 9902
 3rd Qu.:2021-04-29 20:31:53.50   45-54: 3189
 Max.   :2024-09-17 12:03:31.00   55-64:  992
                                  65+  :   94
                                industria       cargo
 other                                :5465   Length:28075
 computing or tech                    :4704   Class :character
 education (higher education)         :2466   Mode  :character
 nonprofits                           :2419
 health care                          :1896
 government and public administration:1890
 (Other)                              :9235
    salario              bonus               moeda       salario_contexto
 Min.   :        0   Min.   :        0   USD    :23387   Length:28075
 1st Qu.:    54000   1st Qu.:        0   CAD    : 1673   Class :character
 Median :    75000   Median :     2000   GBP    : 1591   Mode  :character
 Mean   :   147570   Mean   :    18159   EUR    :  643
 3rd Qu.:   109884   3rd Qu.:    10000   AUD/NZD:  504
 Max.   :870000000   Max.   :120000000   Other  :  160
                     NA's   :7302        (Other):  117
          pais       anos_exp_total anos_exp_area
 U.S.A.     :23606   11-20  :9625   11-20  :6537
 Canada     : 1686   8-10   :5379   5-7    :6520
 U.K.       : 1593   5-7    :4883   2-4    :6252
 other      :  297   21-30  :3637   8-10   :4982
 Germany    :  195   2-4    :3030   21-30  :1868
 New Zealand:  126   31-40  : 870   -1     :1492
 (Other)    :  572   (Other): 651   (Other): 424
                               educacao         sexo
 not informed                      :  222   Length:28075
 High School                       :  639   Class :character
 Some college                      : 2068   Mode  :character
 College degree                    :13527
 Master's degree                   : 8869
 Professional degree (MD, JD, etc.): 1324
 PhD                               : 1426
    etnia
 Length:28075
 Class :character
 Mode  :character
```

sexo

```
s <- sal$sexo
```

```
unique(s)
```

1. 'Woman'
2. 'Non-binary'
3. 'Man'
4. ''
5. 'Other or prefer not to answer'
6. 'Prefer not to answer'

```
s[s == "Woman"] <- "feminino"

s[s == "Man"]   <- "masculino"

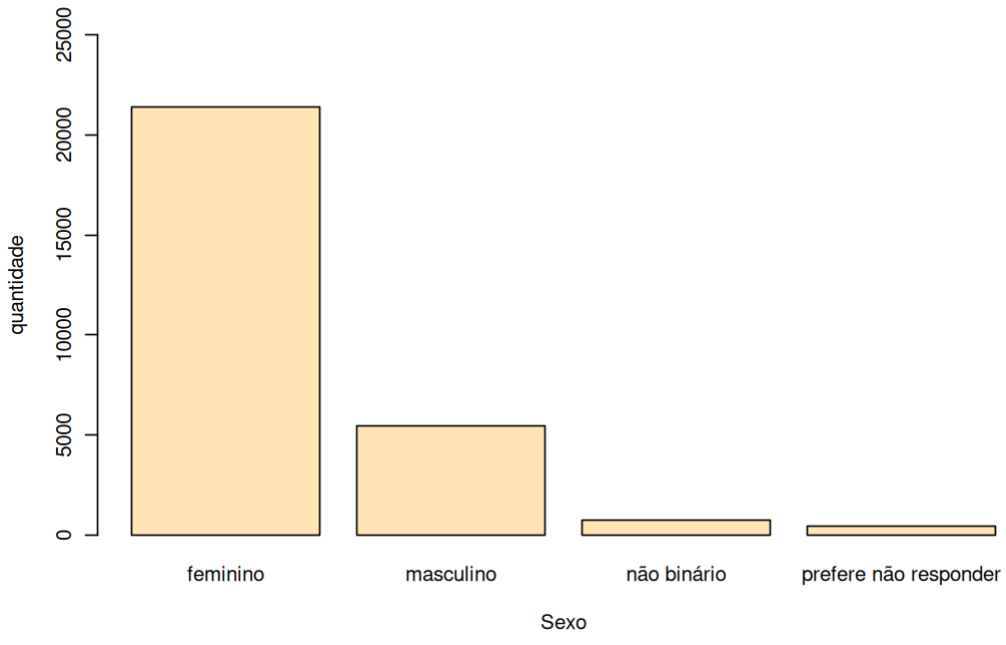
s[s == "Non-binary"]   <- "não binário"

s[s == "Other or prefer not to answer" |
  s == "Prefer not to answer" |
  s == ""]   <- "prefere não responder"
ordem <- c("feminino", "masculino", "não binário",
           "prefere não responder")
s <- factor(s, ordered = TRUE, levels = ordem)
```

```
print(summary(s))
```

```
            feminino            masculino          não binário
               21372                 5488                  745
prefere não responder
                 470
```

```
dimgraf(9,6)
barplot(table(s),ylim=c(0,25000), col="moccasin",
        xlab="Sexo", ylab="quantidade")
```



```
sal$sexo <- s
```

```
summary(sal)
```

```
     tempo                          idade
 Min.   :2021-04-27 11:02:10.00   -18  :   14
 1st Qu.:2021-04-27 14:21:35.50   18-24: 1224
 Median :2021-04-28 13:57:11.00   25-34:12660
 Mean   :2021-05-15 10:18:53.21   35-44: 9902
 3rd Qu.:2021-04-29 20:31:53.50   45-54: 3189
 Max.   :2024-09-17 12:03:31.00   55-64:  992
                                  65+  :   94
                                industria       cargo
 other                               :5465   Length:28075
 computing or tech                   :4704   Class :character
 education (higher education)        :2466   Mode  :character
 nonprofits                          :2419
 health care                         :1896
 government and public administration:1890
 (Other)                             :9235
    salario              bonus               moeda       salario_contexto
 Min.   :        0   Min.   :        0   USD    :23387   Length:28075
 1st Qu.:    54000   1st Qu.:        0   CAD    : 1673   Class :character
 Median :    75000   Median :     2000   GBP    : 1591   Mode  :character
 Mean   :   147570   Mean   :    18159   EUR    :  643
 3rd Qu.:   109884   3rd Qu.:    10000   AUD/NZD:  504
 Max.   :870000000   Max.   :120000000   Other  :  160
                     NA's   :7302        (Other):  117
          pais       anos_exp_total anos_exp_area
 U.S.A.     :23606   11-20  :9625   11-20  :6537
 Canada     : 1686   8-10   :5379   5-7    :6520
 U.K.       : 1593   5-7    :4883   2-4    :6252
 other      :  297   21-30  :3637   8-10   :4982
 Germany    :  195   2-4    :3030   21-30  :1868
 New Zealand:  126   31-40  : 870   -1     :1492
 (Other)    :  572   (Other): 651   (Other): 424
                               educacao                         sexo
 not informed                      :  222   feminino              :21372
 High School                       :  639   masculino             : 5488
 Some college                      : 2068   não binário           :  745
 College degree                    :13527   prefere não responder:  470
 Master's degree                   : 8869
 Professional degree (MD, JD, etc.): 1324
 PhD                               : 1426
    etnia
 Length:28075
 Class :character
 Mode  :character
```

etnia

```r
et <- sal$etnia
```

```r
unique(et)
```

1. 'White'
2. 'Hispanic, Latino, or Spanish origin, White'
3. 'Asian or Asian American, White'
4. 'Asian or Asian American'
5. 'Another option not listed here or prefer not to answer'
6. 'Hispanic, Latino, or Spanish origin'
7. 'Middle Eastern or Northern African'
8. 'Hispanic, Latino, or Spanish origin, Middle Eastern or Northern African, White'
9. 'Black or African American'
10. 'Black or African American, White'
11. ' '
12. 'Black or African American, Hispanic, Latino, or Spanish origin, White'
13. 'Native American or Alaska Native'
14. 'Native American or Alaska Native, White'
15. 'Hispanic, Latino, or Spanish origin, Another option not listed here or prefer not to answer'
16. 'Black or African American, Middle Eastern or Northern African, Native American or Alaska Native, White'
17. 'White, Another option not listed here or prefer not to answer'
18. 'Black or African American, Native American or Alaska Native, White'
19. 'Asian or Asian American, Another option not listed here or prefer not to answer'
20. 'Middle Eastern or Northern African, White'
21. 'Asian or Asian American, Black or African American, White'
22. 'Black or African American, Hispanic, Latino, or Spanish origin'
23. 'Asian or Asian American, Black or African American'
24. 'Asian or Asian American, Hispanic, Latino, or Spanish origin, White'
25. 'Native American or Alaska Native, White, Another option not listed here or prefer not to answer'
26. 'Asian or Asian American, Hispanic, Latino, or Spanish origin'
27. 'Asian or Asian American, Native American or Alaska Native, White'
28. 'Hispanic, Latino, or Spanish origin, Native American or Alaska Native'
29. 'Black or African American, Middle Eastern or Northern African, White'
30. 'Black or African American, Hispanic, Latino, or Spanish origin, Native American or Alaska Native, White'
31. 'Black or African American, Another option not listed here or prefer not to answer'
32. 'Native American or Alaska Native, Another option not listed here or prefer not to answer'
33. 'Asian or Asian American, White, Another option not listed here or prefer not to answer'
34. 'Asian or Asian American, Middle Eastern or Northern African'
35. 'Asian or Asian American, Hispanic, Latino, or Spanish origin, Native American or Alaska Native, White'
36. 'Hispanic, Latino, or Spanish origin, Middle Eastern or Northern African'
37. 'Hispanic, Latino, or Spanish origin, Native American or Alaska Native, White'
38. 'Middle Eastern or Northern African, White, Another option not listed here or prefer not to answer'
39. 'Hispanic, Latino, or Spanish origin, White, Another option not listed here or prefer not to answer'
40. 'Asian or Asian American, Black or African American, Hispanic, Latino, or Spanish origin'
41. 'Asian or Asian American, Black or African American, Native American or Alaska Native, White'
42. 'Middle Eastern or Northern African, Native American or Alaska Native, White'
43. 'Asian or Asian American, Middle Eastern or Northern African, White'
44. 'Black or African American, Middle Eastern or Northern African'
45. 'Hispanic, Latino, or Spanish origin, Native American or Alaska Native, Another option not listed here or prefer not to answer'
46. 'Asian or Asian American, Native American or Alaska Native'
47. 'Middle Eastern or Northern African, Native American or Alaska Native'
48. 'Asian or Asian American, Hispanic, Latino, or Spanish origin, Another option not listed here or prefer not to answer'

49. ‘Asian or Asian American, Hispanic, Latino, or Spanish origin, White, Another option not listed here or prefer not to answer’
50. ‘Asian or Asian American, Black or African American, Hispanic, Latino, or Spanish origin, Middle Eastern or Northern African, Native American or Alaska Native, White, Another option not listed here or prefer not to answer’
51. ‘Black or African American, Native American or Alaska Native’

```
summary(factor(et))
```

1 177Another option not listed here or prefer not to answer
623Asian or Asian American
1406Asian or Asian American, Another option not listed here or prefer not to answer
11Asian or Asian American, Black or African American
9Asian or Asian American, Black or African American, Hispanic, Latino, or Spanish origin
1Asian or Asian American, Black or African American, Hispanic, Latino, or Spanish origin, Middle Eastern or Northern African, Native American or Alaska Native, White, Another option not listed here or prefer not to answer
1Asian or Asian American, Black or African American, Native American or Alaska Native, White
1Asian or Asian American, Black or African American, White
6Asian or Asian American, Hispanic, Latino, or Spanish origin
14Asian or Asian American, Hispanic, Latino, or Spanish origin, Another option not listed here or prefer not to answer
1Asian or Asian American, Hispanic, Latino, or Spanish origin, Native American or Alaska Native, White
1Asian or Asian American, Hispanic, Latino, or Spanish origin, White
14Asian or Asian American, Hispanic, Latino, or Spanish origin, White, Another option not listed here or prefer not to answer
1Asian or Asian American, Middle Eastern or Northern African
7Asian or Asian American, Middle Eastern or Northern African, White
1Asian or Asian American, Native American or Alaska Native
1Asian or Asian American, Native American or Alaska Native, White
4Asian or Asian American, White
343Asian or Asian American, White, Another option not listed here or prefer not to answer
6Black or African American
690Black or African American, Another option not listed here or prefer not to answer
2Black or African American, Hispanic, Latino, or Spanish origin
28Black or African American, Hispanic, Latino, or Spanish origin, Native American or Alaska Native, White
3Black or African American, Hispanic, Latino, or Spanish origin, White
11Black or African American, Middle Eastern or Northern African
2Black or African American, Middle Eastern or Northern African, Native American or Alaska Native, White
1Black or African American, Middle Eastern or Northern African, White
5Black or African American, Native American or Alaska Native
1Black or African American, Native American or Alaska Native, White
6Black or African American, White
123Hispanic, Latino, or Spanish origin
604Hispanic, Latino, or Spanish origin, Another option not listed here or prefer not to answer
6Hispanic, Latino, or Spanish origin, Middle Eastern or Northern African
2Hispanic, Latino, or Spanish origin, Middle Eastern or Northern African, White
5Hispanic, Latino, or Spanish origin, Native American or Alaska Native
10Hispanic, Latino, or Spanish origin, Native American or Alaska Native, Another option not listed here or prefer not to answer
1Hispanic, Latino, or Spanish origin, Native American or Alaska Native, White
10Hispanic, Latino, or Spanish origin, White
384Hispanic, Latino, or Spanish origin, White, Another option not listed here or prefer not to answer
1Middle Eastern or Northern African
69Middle Eastern or Northern African, Native American or Alaska Native

1Middle Eastern or Northern African, Native American or Alaska Native, White
1Middle Eastern or Northern African, White
83Middle Eastern or Northern African, White, Another option not listed here or prefer not to answer
2Native American or Alaska Native
43Native American or Alaska Native, Another option not listed here or prefer not to answer
1Native American or Alaska Native, White
67Native American or Alaska Native, White, Another option not listed here or prefer not to answer
1White
23219White, Another option not listed here or prefer not to answer
65

```
et[grepl("black", et, ignore.case = TRUE) & grepl("white", et, ignore.case = TRUE)] <- 0 # "multiple"
et[grepl("black", et, ignore.case = TRUE) & grepl("asian", et, ignore.case = TRUE)] <- 0 # "multiple"
et[grepl("black", et, ignore.case = TRUE) & grepl("northern", et, ignore.case = TRUE)] <- 0 # "multiple"
et[grepl("black", et, ignore.case = TRUE) & grepl("hispanic", et, ignore.case = TRUE)] <- 0 # "multiple"
et[grepl("black", et, ignore.case = TRUE) & grepl("native", et, ignore.case = TRUE)] <- 0 # "multiple"
et[grepl("asian", et, ignore.case = TRUE) & grepl("northern", et, ignore.case = TRUE)] <- 0 # "multiple"
et[grepl("asian", et, ignore.case = TRUE) & grepl("latino", et, ignore.case = TRUE)] <- 0 # "multiple"
et[grepl("asian", et, ignore.case = TRUE) & grepl("native", et, ignore.case = TRUE)] <- 0 # "multiple"
et[grepl("latino", et, ignore.case = TRUE) & grepl("northern", et, ignore.case = TRUE)] <- 0 # "multiple"
et[grepl("latino", et, ignore.case = TRUE) & grepl("native", et, ignore.case = TRUE)] <- 0 # "multiple"
et[grepl("northern", et, ignore.case = TRUE) & grepl("native", et, ignore.case = TRUE)] <- 0 # "multiple"
```

```
et[grepl("prefer not to answer",et,ignore.case = TRUE)] <- "1" # "prefer not to answer"
et[et == ""] <- '1' # "prefer not to answer"
```

```
et[et == 'Asian or Asian American'] <- '2' # Asian or Asian American
et[et == 'Asian or Asian American, White'] <- '2'
```

```
et[et == 'Hispanic, Latino, or Spanish origin'] <- '3' #Hispanic, Latino, or Spanish origin
et[et == 'Hispanic, Latino, or Spanish origin, White'] <- '3'
```

```
et[et == 'Native American or Alaska Native'] <- '4' #Native American or Alaska Native
et[et == 'Native American or Alaska Native, White'] <- '4' #Native American or Alaska Native
```

```
et[et == 'Black or African American'] <- '5' #Black or African American
```

```
et[et == 'Middle Eastern or Northern African'] <- '6' #Middle Eastern or Northern African
et[et == 'Middle Eastern or Northern African, White'] <- '6'
```

```
et[et == 'White'] <- 'Branco'
et[et == '0'] <- 'Múltiplo'
et[et == '1'] <- 'Não forneceu'
et[et == '2'] <- 'Asiático'
et[et == '3'] <- 'Latino/Hispânico'
et[et == '4'] <- 'Orig. América do Norte'
et[et == '5'] <- 'Afrodescendente'
et[et == '6'] <- 'Oriente médio'
ordem <- c('Afrodescendente', 'Asiático', 'Branco', 'Latino/Hispânico', 'Oriente médio',
           'Orig. América do Norte', 'Múltiplo', 'Não forneceu')
```

```
et <- factor(et, ordered = TRUE, levels = ordem)
```

```
print(summary(et))
```

```
      Afrodescendente               Asiático                 Branco
                  690                   1749                  23219
     Latino/Hispânico          Oriente médio Orig. América do Norte
                  988                    152                    110
             Múltiplo           Não forneceu
                  272                    895
```

```
sal$etnia <- et
```

```
print(summary(sal))
```

```
    tempo                         idade
 Min.   :2021-04-27 11:02:10.00   -18  :   14
 1st Qu.:2021-04-27 14:21:35.50   18-24: 1224
 Median :2021-04-28 13:57:11.00   25-34:12660
 Mean   :2021-05-15 10:18:53.21   35-44: 9902
 3rd Qu.:2021-04-29 20:31:53.50   45-54: 3189
 Max.   :2024-09-17 12:03:31.00   55-64:  992
                                  65+  :   94
                               industria       cargo
 other                              :5465   Length:28075
 computing or tech                  :4704   Class :character
 education (higher education)       :2466   Mode  :character
 nonprofits                         :2419
 health care                        :1896
 government and public administration:1890
 (Other)                            :9235
    salario              bonus               moeda       salario_contexto
 Min.   :        0   Min.   :        0   USD    :23387   Length:28075
 1st Qu.:    54000   1st Qu.:        0   CAD    : 1673   Class :character
 Median :    75000   Median :     2000   GBP    : 1591   Mode  :character
 Mean   :   147570   Mean   :    18159   EUR    :  643
 3rd Qu.:   109884   3rd Qu.:    10000   AUD/NZD:  504
 Max.   :870000000   Max.   :120000000   Other  :  160
                     NA's   :7302        (Other):  117
          pais       anos_exp_total anos_exp_area
 U.S.A.     :23606   11-20  :9625   11-20  :6537
 Canada     : 1686   8-10   :5379   5-7    :6520
 U.K.       : 1593   5-7    :4883   2-4    :6252
 other      :  297   21-30  :3637   8-10   :4982
 Germany    :  195   2-4    :3030   21-30  :1868
 New Zealand:  126   31-40  : 870   -1     :1492
 (Other)    :  572   (Other): 651   (Other): 424
                               educacao                         sexo
 not informed                      :  222   feminino             :21372
 High School                       :  639   masculino            : 5488
 Some college                      : 2068   não binário          :  745
 College degree                    :13527   prefere não responder:  470
 Master's degree                   : 8869
 Professional degree (MD, JD, etc.): 1324
 PhD                               : 1426
             etnia
 Branco          :23219
```

```
 Asiático        : 1749
 Latino/Hispânico:  988
 Não forneceu    :  895
 Afrodescendente :  690
 Múltiplo        :  272
 (Other)         :  262
```

```
save(sal, file="SalarySurvey2021_02_limpeza.RData")
```

```
rm(list=ls())
dimgraf <- function (x=11, y=7) { options(repr.plot.width=x,repr.plot.height=y) }
load("SalarySurvey2021_02_limpeza.RData")
ls()
```

1. 'dimgraf'
2. 'sal'

1.4 Conformação do salário em USD

```
s <- sal$salario
nomes <- names(sal)
nomes[nomes == 'salario'] <- 'salario_antigo'
names(sal) <- nomes
m <- as.character(sal$moeda)
```

Pela taxa de câmbio de 28/09/24:

```
# (AUD = 0,69 + NZD = 0,63)/2 -> 0,66
s <- ifelse(m == 'AUD/NZD', s * 0.66, s)
s <- ifelse(m == 'CAD', s *  0.74, s)
s <- ifelse(m == 'CHF', s *  1.19, s)
s <- ifelse(m == 'EUR', s *  1.12, s)
s <- ifelse(m == 'GBP', s *  1.34, s)
s <- ifelse(m == 'HKD', s *  0.13, s)
s <- ifelse(m == 'JPY', s * 0.007, s)
s <- ifelse(m == 'SEK', s *  0.13, s)
s <- ifelse(m == 'ZAR', s *  0.06, s)
```

```
s[m == 'Other'] <- NA
```

```
print(summary(factor(m)))
```

```
AUD/NZD     CAD     CHF     EUR     GBP     HKD     JPY   Other     SEK     USD
    504    1673      37     643    1591       4      23     160      37   23387
    ZAR
     16
```

```
summary(s)
```

```
     Min.   1st Qu.    Median      Mean   3rd Qu.      Max.      NA's
        0     53280     74000     91970    106000 102000000       160
```

```
sal$salario <- s
```

Quais são essses salários que parecem exagerados?

```
sal[!is.na(sal$salario) &
    (sal$salario > 200000) &
    sal$moeda != 'USD' ,c('pais','salario_antigo',
                          'salario_contexto','moeda', 'salario') ]
```

A data.frame: 45 × 5

	pais <fct>	salario_antigo <dbl>	salario_contexto <chr>	moeda <fct>	salario <dbl>
266	Germany	260000		EUR	291200
3010	Canada	600000		CAD	444000
5582	U.K.	910000	Bonus depends on a LOT. With covid, it took a hit to about 150, but in all other years it's been 250	GBP	1219400
8588	U.K.	160000		GBP	214400
9929	U.K.	156000	I receive about 100k in equity per year as well as a performance bonus.	GBP	209040
11658	Norway	550000		EUR	616000
11755	U.K.	260000		GBP	348400
11792	U.K.	150000		GBP	201000
12437	U.K.	185000		GBP	247900
12556	U.S.A.	400000		AUD/NZD	264000
12570	U.K.	190000		GBP	254600
12607	U.K.	175000		GBP	234500
12723	U.K.	200000	All food and a house come as part of the salary as work in a boarding school. Salary is for the full year's work but I have 18 weeks holiday out of term time, which makes the salary much more justifiable than it first appears.	GBP	268000
13497	Denmark	444000		EUR	497280
14073	Norway	689000		GBP	923260

	pais <fct>	salario_antigo <dbl>	salario_contexto <chr>	moeda <fct>	salario <dbl>
14208	U.K.	950000	Additional compensation is a profit share	GBP	1273000
14298	U.K.	170000		GBP	227800
14425	France	350000		EUR	392000
14430	Sweden	2300000		SEK	299000
14532	France	207000		EUR	231840
14685	U.K.	157000		GBP	210380
14979	Switzerland	170000		CHF	202300
15439	Switzerland	200000		CHF	238000
15818	U.K.	250000		GBP	335000
15983	U.K.	150000		GBP	201000
16151	Germany	850000		EUR	952000
17537	Germany	360000		EUR	403200
18766	U.K.	218000		GBP	292120
18773	U.K.	200000		GBP	268000
18803	Germany	290000	Including stock	EUR	324800
18921	U.K.	176000		GBP	235840
19321	Ireland	180000		EUR	201600
19471	U.K.	500000		GBP	670000
19855	Switzerland	220000		CHF	261800
20453	U.K.	180000		GBP	241200
21294	India	1000000		CHF	1190000
21342	U.K.	165000		GBP	221100
21684	Canada	350000		CAD	259000
22052	U.K.	440000		GBP	589600
23331	U.K.	180000		GBP	241200
23560	Canada	960000		CAD	710400
23851	Spain	214000		EUR	239680
25005	Belgium	180000		EUR	201600
25838	U.K.	190000		GBP	254600
26395	U.S.A.	1000000		AUD/NZD	660000

```
sal[!is.na(sal$salario) &
    (sal$salario < 1200) &                      # menos que 100 USD por mês
    sal$moeda != 'USD' ,c('pais','salario_antigo',
                          'salario_contexto','moeda', 'salario') ]
```

A data.frame: 33 × 5

	pais <fct>	salario_antigo <dbl>	salario_contexto <chr>	moeda <fct>	salario <dbl>
167	Finland	35		EUR	39.20
2230	Germany	33	Before tax, insurance etc.	EUR	36.96
4547	Germany	63		EUR	70.56
5140	Canada	118		CAD	87.32
5880	other	18		EUR	20.16
6295	U.K.	50		GBP	67.00
6930	Canada	52		CAD	38.48
8593	Germany	100		EUR	112.00
8680	Canada	91		CAD	67.34
8817	Germany	80		EUR	89.60

	pais <fct>	salario_antigo <dbl>	salario_contexto <chr>	moeda <fct>	salario <dbl>
10088	U.K.	65		GBP	87.10
11465	Germany	55		EUR	61.60
11703	Italy	40		EUR	44.80
11915	Spain	30		EUR	33.60
12909	other	36	The additional monetary compensations are vacation pay (netto) and extra's for helping out within the company and a thank-you-for-surviving-Covid-/-the-end-of-the-world-pay	EUR	40.32
13315	Japan	27500		JPY	192.50
14183	Canada	70		CAD	51.80
14299	France	37		EUR	41.44
14311	France	37		EUR	41.44
14874	Germany	65	Paid 35 days of vacation and 13. monthly salary	EUR	72.80
15736	U.K.	28		GBP	37.52
17101	Ireland	36		EUR	40.32
18619	Germany	100	Stock options are part of salsry	EUR	112.00
18845	other	60		EUR	67.20
18877	U.K.	51		GBP	68.34
19296	Germany	50		EUR	56.00
21360	Spain	51		EUR	57.12
26188	Canada	90		CAD	66.60
26537	U.K.	40		GBP	53.60
27731	Spain	50		ZAR	3.00
28016	India	0	hvhv	GBP	0.00
28027	Germany	70		EUR	78.40
28050	other	5	ff	EUR	5.60

Claramente salários anuais menores que 10 USD é um erro. Eliminando:

```
sal <- sal[sal$salario > 10,]
```

Para os demais casos, parece que o respondente considerou o salário * 1000, ou seja, quando encontramos 70, provavelmente seria 70k (70000). Porém, como é apenas especulação, esses valores também serão excluídos por estarem incorretos frente aos demais:

```
summary(sal$salario)
```

```
     Min.   1st Qu.    Median      Mean   3rd Qu.      Max.      NA's
       15     53280     74400     92043    106000 102000000       160
```

```
sal <- sal[sal$salario >= 1200,]
```

```
summary(sal$salario)
```

```
   Min.   1st Qu.    Median      Mean   3rd Qu.      Max.      NA's
   1200     53600     75000     92344    106000 102000000       160
```

```
save(sal, file="SalarySurvey2021_03_completo.RData")
```

1.5 Separando observações próprias para análise

```
rm(list=ls())
dimgraf <- function (x=11, y=7) { options(repr.plot.width=x,repr.plot.height=y) }
load("SalarySurvey2021_03_completo.RData")
ls()
```

1. 'dimgraf'
2. 'sal'

```
str(sal)
```

```
'data.frame':   27962 obs. of  15 variables:
 $ tempo           : POSIXct, format: "2021-04-27 11:02:10" "2021-04-27 11:02:22" ...
 $ idade           : Ord.factor w/ 7 levels "-18"<"18-24"<..: 3 3 3 3 3 3 3 3 5 4 ...
 $ industria       : Factor w/ 15 levels "accounting, banking & finance",..: 4 3 1 13 1 4 14 5 3 1 ...
 $ cargo           : chr  "Research and Instruction Librarian" "Change & Internal Communications Manager" "Marketi
 $ salario_antigo  : num  55000 54600 34000 62000 60000 62000 33000 50000 112000 45000 ...
 $ bonus           : int  0 4000 NA 3000 7000 NA 2000 NA 10000 0 ...
 $ moeda           : Factor w/ 11 levels "AUD/NZD","CAD",..: 10 5 10 10 10 10 10 10 10 10 ...
 $ salario_contexto: chr  "" "" "" "" ...
 $ pais            : Factor w/ 25 levels "Belgium","Brazil",..: 25 24 25 25 25 25 25 25 25 25 ...
 $ anos_exp_total  : Ord.factor w/ 8 levels "-1"<"2-4"<"5-7"<..: 3 4 2 4 4 4 2 3 6 6 ...
 $ anos_exp_area   : Ord.factor w/ 8 levels "-1"<"2-4"<"5-7"<..: 3 3 2 3 3 2 2 3 6 6 ...
 $ educacao        : Ord.factor w/ 7 levels "not informed"<..: 5 4 4 4 4 5 4 5 4 4 ...
 $ sexo            : Ord.factor w/ 4 levels "feminino"<"masculino"<..: 1 3 1 1 1 2 1 2 1 1 ...
 $ etnia           : Ord.factor w/ 8 levels "Afrodescendente"<..: 3 3 3 3 3 3 3 3 3 4 ...
 $ salario         : num  55000 73164 34000 62000 60000 ...
```

Uma vez que o salário está padronizado em USD, moeda, contexto, bônus, salário antigo podem ser excluídos pois não são atributos interessantes para a análise.

```
sal$moeda <- NULL
sal$salario_contexto <- NULL
sal$salario_antigo <- NULL
sal$bonus <- NULL
```

O marco temporal não é particularmente útil, pois o salário não é suposto flutuar por minutos ou dias. Assim, talvez obter apenas o ano e o mês.

```
sal$ano <- as.factor(as.integer(format(sal$tempo, "%Y")))
sal$mes <- as.factor(as.integer(format(sal$tempo, "%m")))
```

```
print(summary(sal$ano))
print(summary(sal$mes))
sal$tempo <- NULL
```

```
 2021  2022  2023  2024  NA's
27362   307    73    60   160
    1     2     3     4     5     6     7     8     9    10    11    12  NA's
   58    67    53 22655  3619   152   100   162    76   697    96    67   160
```

As observações sem salário (NA) podem ser interessantes para avaliar um modelo, mas não para treinar ou analisar o comportamento do salário. Será salvo então um conjunto de dados com os NAs e depois serão descartados.

```
summary(sal)
```

```
     idade                                industria        cargo
 25-34  :12536   other                         : 5410   Length:27962
 35-44  : 9833   computing or tech             : 4634   Class :character
 45-54  : 3153   education (higher education): 2450   Mode  :character
 18-24  : 1197   nonprofits                    : 2407
 55-64  :  977   health care                   : 1881
 (Other):  106   (Other)                       :11020
 NA's   :  160   NA's                          :  160
      pais       anos_exp_total anos_exp_area
 U.S.A. :23516   11-20  :9554   11-20  :6489
 Canada : 1679   8-10   :5341   5-7    :6470
 U.K.   : 1586   5-7    :4844   2-4    :6174
 other  :  245   21-30  :3601   8-10   :4939
 Germany:  186   2-4    :2971   21-30  :1848
 (Other):  590   (Other):1491   (Other):1882
 NA's   :  160   NA's   : 160   NA's   : 160
                                educacao                          sexo
 College degree                    :13416   feminino             :21201
 Master's degree                   : 8791   masculino            : 5405
 Some college                      : 2032   não binário          :  740
 PhD                               : 1416   prefere não responder:  456
 Professional degree (MD, JD, etc.): 1311   NA's                 :  160
 (Other)                           :  836
 NA's                              :  160
              etnia          salario            ano             mes
 Branco          :23074   Min.   :     1200   2021:27362   4      :22655
 Asiático        : 1697   1st Qu.:    53600   2022:  307   5      : 3619
 Latino/Hispânico:  970   Median :    75000   2023:   73   10     :  697
 Não forneceu    :  863   Mean   :    92344   2024:   60   8      :  162
 Afrodescendente :  677   3rd Qu.:   106000   NA's:  160   6      :  152
 (Other)         :  521   Max.   :102000000                (Other):  517
 NA's            :  160   NA's   :160                      NA's   :  160
```

```
sal <- sal[!is.na(sal$salario),]
```

```
summary(sal)
```

```
   idade                                    industria       cargo
 -18  :   12   other                             :5410   Length:27802
```

```
18-24: 1197   computing or tech                  :4634   Class :character
25-34:12536   education (higher education)       :2450   Mode  :character
35-44: 9833   nonprofits                         :2407
45-54: 3153   health care                        :1881
55-64:  977   government and public administration:1877
65+  :   94   (Other)                           :9143
         pais       anos_exp_total anos_exp_area
U.S.A.     :23516   11-20  :9554   11-20  :6489
Canada     : 1679   8-10   :5341   5-7    :6470
U.K.       : 1586   5-7    :4844   2-4    :6174
other      :  245   21-30  :3601   8-10   :4939
Germany    :  186   2-4    :2971   21-30  :1848
New Zealand:  124   31-40  : 858   -1     :1464
(Other)    :  466   (Other): 633   (Other): 418
                            educacao                        sexo
not informed                    :  208   feminino             :21201
High School                     :  628   masculino            : 5405
Some college                    : 2032   não binário          :  740
College degree                  :13416   prefere não responder:  456
Master's degree                 : 8791
Professional degree (MD, JD, etc.): 1311
PhD                             : 1416
             etnia          salario            ano             mes
Branco          :23074   Min.   :     1200   2021:27362   4      :22655
Asiático        : 1697   1st Qu.:    53600   2022:  307   5      : 3619
Latino/Hispânico:  970   Median :    75000   2023:   73   10     :  697
Não forneceu    :  863   Mean   :    92344   2024:   60   8      :  162
Afrodescendente :  677   3rd Qu.:   106000                6      :  152
Múltiplo        :  269   Max.   :102000000                7      :  100
(Other)         :  252                                    (Other):  417
```

```
save(sal, file="SalarySurvey2021_04_sem_NA.RData")
```